

Chapter 3: Linear Maps

Linear Algebra Done Right

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3.A The Vector Space of Linear Maps

Problem (1)

Suppose $b, c \in \mathbb{R}$. Define $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ by

$$T(x, y, z) = (2x - 4y + 3z + b, 6x + cxyz).$$

Show that T is linear if and only if $b = c = 0$.

In a linear map, all of the terms must be linear, otherwise, when you add two elements of a linear map, then their results won't be an element of the output and thus it won't be a linear map. Therefore, the two non-linears, b and $cxyz$, must equal 0, so $b = c = 0$, as desired.

Problem (2)

Suppose $b, c \in \mathbb{R}$. Define $T : \mathcal{P}(\mathbb{R}) \rightarrow \mathbb{R}^2$ by

$$Tp = \left(3p(4) + 5p'(6) + bp(1)p(2), \int_{-1}^2 x^3 p(x) dx + c \sin p(0) \right).$$

Show that T is linear if and only if $b = c = 0$.

In a similar reasoning as the problem above, if there are any nonlinear terms, meaning that it has to be a polynomial by itself in this case, since $bp(1)p(2) + bq(1)q(2) \neq b(p+q)(1) \cdot (p+q)(2)$. Similarly, $c \sin p(0) + c \sin q(0) \neq c \sin(p+q)(0)$, so we're done.

Problem (3)

Suppose $T \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$. Show that there exist scalars $A_{j,k} \in \mathbb{F}$ for $j = 1, \dots, m$ and $k = 1, \dots, n$ such that

$$T(x_1, \dots, x_n) = (A_{1,1}x_1 + \dots + A_{1,n}x_n, \dots, A_{m,1}x_1 + \dots + A_{m,n}x_n)$$

for every $(x_1, \dots, x_n) \in \mathbb{F}^n$.

[The exercise above shows that T has the form promised in the last item of Example 3.4.]

Since all these scalars are independent of each other, then we have a matrix. We want to show that if the output is, say, $(y_1, \dots, y_m)$, then there exists $A_{j,k} \in \mathbb{F}$ such that

$$\begin{aligned} A_{1,1}x_1 + \dots + A_{1,n}x_n &= y_1 \\ &\vdots \\ A_{m,1}x_1 + \dots + A_{m,n}x_n &= y_m. \end{aligned}$$

Since there are n variables and m equations, we can make up values of $A_{j,k}$ such that all of the equations hold.

Problem (4)

Suppose $T \in \mathcal{L}(V, W)$ and $v_1, \dots, v_m$ is a list of vectors in V such that $Tv_1, \dots, Tv_m$ is a linearly independent list in W . Prove that $v_1, \dots, v_m$ is linearly independent.

Since $Tv_1, \dots, Tv_m$ are linearly independent, then none of the elements are combinations of any of the other elements. Since $T(v_1 + v_2) = Tv_1 + Tv_2$, then this also means that none of the elements of $v_1, \dots, v_m$ are combinations of any of the other elements, so they are also linearly independent, as desired.

Problem (5)

Prove the assertion in 3.7.

To prove that $\mathcal{L}(V, W)$ is a vector space, we need to prove that addition and multiplication hold for the set of linear maps. So basically, the set of linear maps is a linear map. Hehe. Anyway consider two linear maps, L_1 and L_2 . Then note that we need to prove that $L_1 + L_2$ and λL_1 are also linear maps. To prove that $L_1 + L_2$ is a linear map, consider two vectors v_1 and v_2 . We want to show that $(L_1 + L_2)(v_1 + v_2) = (L_1 + L_2)v_1 + (L_1 + L_2)v_2$.

$$\begin{aligned} (L_1 + L_2)(v_1 + v_2) &= L_1(v_1 + v_2) + L_2(v_1 + v_2) \\ &= L_1v_1 + L_1v_2 + L_2v_1 + L_2v_2 \\ &= (L_1v_1 + L_2v_1) + (L_1v_2 + L_2v_2) \\ &= (L_1 + L_2)v_1 + (L_1 + L_2)v_2, \end{aligned}$$

since L_1 and L_2 are linear maps. Also,

$$(L_1 + L_2)(\lambda v_1) = L_1(\lambda v_1) + L_2(\lambda v_1) = \lambda L_1v_1 + \lambda L_2v_1 = \lambda(L_1v_1 + L_2v_1) = \lambda(L_1 + L_2)(v_1)$$

so $L_1 + L_2$ is a linear map, as desired.

We now show that λL is a linear map.

$$(\lambda L)(v_1 + v_2) = \lambda(L(v_1 + v_2)) = \lambda(Lv_1 + Lv_2) = (\lambda L)v_1 + (\lambda L)v_2$$

so it satisfies summation. Now

$$(\lambda_1 L)(\lambda_2 v) = \lambda_1(L(\lambda_2 v)) = \lambda_1(\lambda_2(Lv)) = \lambda_1\lambda_2(Lv) = \lambda_2((\lambda_1 L)v),$$

as desired. Because L is a linear map itself.

Problem (6)

Prove the assertion in 3.9.

First assertion: $(T_1T_2)T_3 = T_1(T_2T_3)$. We want to show that $(T_1v \cdot T_2v) \cdot T_3v = T_1v \cdot (T_2v \cdot T_3v)$. Since this holds for vectors, we're done.

Second assertion. Obviously if we multiply anything by its identity, depending on its direction, we will still get T .

Third assertion. $(S_1 + S_2)Tv = S_1Tv + S_2Tv$, so $(S_1 + S_2)T = S_1T + S_2T$ and $S(T_1v + T_2v) = ST_1v + ST_2v$, so $S(T_1 + T_2) = ST_1 + ST_2$.

Problem (7)

Show that every linear map from a 1-dimensional vector space to itself is multiplication by some scalar. More precisely, prove that if $\dim V = 1$ and $T \in \mathcal{L}(V, V)$, then there exists $\lambda \in \mathbb{F}$ such that $Tv = \lambda v$ for all $v \in V$.

8 If we have $v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ and let $\varphi(v) = \sqrt{v_1v_2}$. Obviously if we have $u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$, then $\varphi(u + v) \neq \varphi(u) + \varphi(v)$, but $\varphi(av) = a\varphi(v)$.

9

14 If we let $S = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $T = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, then $ST = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ and $TS = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$. Clearly $ST \neq TS$.

3.B Null Spaces and Ranges

1 Consider the map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^2$. Then if we map $T(a, b, c, d, e)$ to $(a, b, 0, 0, 0)$, then the range has dimension 2, and the null space consists of all quintuplets of the form $(0, 0, c, d, e)$, which has dimension 3.

2 Suppose $S : X \rightarrow Y$ and $T : U \rightarrow V$. Then if we have $x \in X$, which maps to a $y \in Y$, then $y \in \text{null } T$, so $T(y) = T(S(x)) = 0$, so $TS = 0$, so $(ST)^2 = 0$.

3

(a) Since $(v_1, \dots, v_m)$ encompasses all of V , then the map from the vectors, $\mathbb{F}^m$, to V must be surjective.

(b) If $v_1, \dots, v_m$ are linearly independent, this means that the null space has degree 0, and consists of only $\{0\}$.

4 Since we know that $\dim \text{null } T + \dim \text{range } T = \dim V$ but $\dim V = 5$ and $\dim \text{null } T > 2$ and $\dim \text{range } T = 4$, which is impossible.

5 How about something like $T : (a, b, c, d) \rightarrow (a, b, 0, 0)$. Then the range has dimension 2 but the null space also has dimension 2.

6 Why, you ask? (I got this from All Might). Because we know that $\dim V = \dim \text{range } T + \dim \text{null } T$, but $\dim V = 5$, which is odd, so we can't split it evenly.

7 Observe that since $\dim \text{range } T \geq \dim V$, but $\dim \text{null } T \geq 1$ because T is not injective, then it is impossible for $\dim \text{range } T + \dim \text{null } T$ to equal $\dim V$, as desired.

8 Because T is not surjective, this means that $\dim \text{range } T < \dim W \leq \dim V$ and $\dim \text{null } T$. Then $\dim \text{range } T < \dim W \leq \dim V$. Hence, $\dim \text{null } T + \dim \text{range } T > \dim V$, so we're done.

9 Since $v_1, \dots, v_n$ is linearly independent, then since T is injective, we also know that the null space of T equals 0, so $Tv_1, \dots, Tv_n$ is also linearly independent because we know

that T is a linear map, so $Tv + Tw = T(v + w)$. And since its injective, each v maps to a distinct output, so we can look at the inputs instead, which are linearly independent.

10 Since T is a linear map from V to W and $v_1, \dots, v_n$ spans V this also means that $Tv_1, \dots, Tv_n$ spans W .

11 $(S_1 S_2 \cdots S_n)(v) = (S_1(v))(S_2(v)) \cdots (S_n(v))$. So if we have another w such that $(S_1 S_2 \cdots S_n)(w) = (S_1 S_2 \cdots S_n)(v)$, then we must have $S_1(v) = S_1(w)$, etc, so since $S_1, \dots, S_n$ are injective, then $v = w$.

12 We want to show that T is surjective. Obviously, if T is surjective, then U suffices as the input because we really only need one zero, and the rest would be taken care of if we let U be the set consisting of all inputs that don't result in an output of 0, plus one element from the null set, which is $\{0\}$.

13 Observe that the null space consists of points of the form $(5a, a, 7b, b)$, which has dimension 2. Since $\dim \text{null } T + \dim \text{range } T = \dim V = 4$, then $\dim \text{range } T = 2 = \dim W$, so T is surjective, as desired.

14 Since $\text{null } T = U$, then since U is 3-dimensional, then $\dim \text{null } T = 3$ as well. $\dim \text{null } T + \dim \text{range } T = \dim V = 8$, so $\dim \text{range } T = 5 = \dim W$, so T is surjective, as desired.

15 So the null space consists of quintuplets of the form $(3a, a, b, b, b)$, which has dimension 2. Since $\dim \text{range } T = \dim V - \dim \text{null } T = 5 - 2 = 3$, then $\dim \text{range } T > \dim W$, which is impossible.

16 Thus there exists values such that

$$Tv = \varphi_1(v)w_1 + \cdots + \varphi_m(v)w_m.$$

17 Consider the map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^2$. Then if we map $T(a, b, c, d, e)$ to $(a, b, 0, 0, 0)$, then the range has dimension 2, and the null space consists of all quintuplets of the form $(0, 0, c, d, e)$, which has dimension 3.

18 Suppose $S : X \rightarrow Y$ and $T : U \rightarrow V$. Then if we have $x \in X$, which maps to a $y \in Y$, then $y \in \text{null } T$, so $T(y) = T(S(x)) = 0$, so $TS = 0$, so $(ST)^2 = 0$.

19 Since $(v_1, \dots, v_m)$ encompasses all of V , then the map from the vectors, $\mathbb{F}^m$, to V must be surjective. If $v_1, \dots, v_m$ are linearly independent, this means that the null space has degree 0, and consists of only $\{0\}$.

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25 Since $v_1, \dots, v_n$ is linearly independent, then since T is injective, we also know that the null space of T equals 0, so $Tv_1, \dots, Tv_n$ is also linearly independent because we know that T is a linear map, so $Tv + Tw = T(v + w)$. And since its injective, each v maps to a distinct output, so we can look at the inputs instead, which are linearly independent.

26 Since T is a linear map from V to W and $v_1, \dots, v_n$ spans V this also means that $Tv_1, \dots, Tv_n$ spans W .

27 $(S_1 S_2 \cdots S_n)(v) = (S_1(v))(S_2(v)) \cdots (S_n(v))$. So if we have another w such that $(S_1 S_2 \cdots S_n)(w) = (S_1 S_2 \cdots S_n)(v)$, then we must have $S_1(v) = S_1(w)$, etc, so since $S_1, \dots, S_n$ are injective, then $v = w$.

28 We want to show that T is surjective. Obviously, if T is surjective, then U suffices as the input because we really only need one zero, and the rest would be taken care of if we let U be the set consisting of all inputs that don't result in an output of 0, plus one element from the null set, which is $\{0\}$.

29 Observe that the null space consists of points of the form $(5a, a, 7b, b)$, which has dimension 2. Since $\dim \text{null } T + \dim \text{range } T = \dim V = 4$, then $\dim \text{range } T = 2 = \dim W$, so T is surjective, as desired.

30 Since $\text{null } T = U$, then since U is 3-dimensional, then $\dim \text{null } T = 3$ as well. $\dim \text{null } T + \dim \text{range } T = \dim V = 8$, so $\dim \text{range } T = 5 = \dim W$, so T is surjective, as desired.

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3.C Matrices

1 Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. There exists a basis $w_1, \dots, w_n$ of W such that all the entries in the first column of $\mathcal{M}(T)$ (with respect to the bases $v_1, \dots, v_m$ and $w_1, \dots, w_n$) are 0 except for possibly T as a linear map from V to W such that there is nothing except possibly, a 0 in the first row and first column. If we have the matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

In this case, we have a matrix with a 1's along the diagonal, meaning places where the number of rows and number of columns are equal, and 0 everywhere else.

2 Prove that there exists a basis $v_1, \dots, v_m$

6 If $\dim \text{range } T = 1$, this means that $\dim \text{null } T = \dim V - 1$. Therefore there needs to exist a basis of V and a basis of W such that all entries of $\mathcal{M}(T)$ equal 1, because this would

7 We know that $\mathcal{M}(T)$ is defined as the set of numbers $a_{i,j}$ for $1 \leq i \leq m$, $1 \leq j \leq n$ such that for all j ,

$$w_j = a_{1,j}v_1 + a_{2,j}v_2 + \cdots + a_{m,j}v_m.$$

So, if we add w_{s_j} to w_{t_j} (these are the respective bases of S and T), and let a and b denote the values of the entries in the matrices, we get

$$\mathcal{M}(S + T) = (a_{1,i} + b_{1,i})v_1 + \cdots + (a_{m,i} + b_{m,i})v_m$$

and

$$\begin{aligned} \mathcal{M}(S) + \mathcal{M}(T) &= [a_{1,i}v_1 + \cdots + a_{m,i}v_m] \\ &\quad + [b_{1,i}v_1 + \cdots + b_{m,i}v_m] \\ &= (a_{1,i} + b_{1,i})v_1 + \cdots + (a_{m,i} + b_{m,i})v_m \end{aligned}$$

as desired.

8 If we have

$$Tv_i = a_{1,i}v_1 + \cdots + a_{m,i}v_i,$$

then if we multiply T times λ , we get

$$\lambda a_{1,i}v_1 + \cdots + \lambda a_{m,i}v_i = a_{1,i}(\lambda v_1) + \cdots + a_{m,i}(\lambda v_i)$$

as desired.

9 Observe that we wish to prove that when we multiply a matrix by a vector, we get a combination of all of the previous vectors and put them together. This is clear because we already have a collection of vectors to start with. A single value times matrix put together gives a vector of values times matrix.

12 Consider $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and $C = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$. Then

$$AC = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$$

and

$$CA = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$$

as desired.

14 Observe that if $(AB)C$ makes sense, then if A is $m \times n$, then B must be $n \times p$ for some p , so AB is $m \times p$, so C is $p \times q$ for some q . Thus BC is $n \times q$, so $A(BC)$ is $m \times q$, which is defined.

15 We already know that the entry in any given row and column of A^2 is equal to the sum of the products of $a_{i,j}a_{j,k}$, so continuing with A^3 gives $a_{j,p}a_{p,r}a_{r,k}$.

3.D Invertibility and Isomorphic Vector Spaces

1 Observe that if both T (which is a linear map from U to V) and S which is a linear map from V to W are invertible, then ST , which is a linear map from U to W is invertible, since it's the product of S and T , and thus $ST = T^{-1}S^{-1}$.

2 This doesn't hold because the sum of two noninvertible operators might just be invertible.

3 If there exists an invertible operator $T \in \mathcal{L}(V)$ such that $Tu = Su$ for every $u \in U$, then this must mean that since T is both injective and surjective, then since $Tu = Su$, then so must S . Conversely, if S is injective, then there must exist T such that $Tu = Su$.

4 If $\text{null } T_1 = \text{null } T_2$, this implies that the map from the output of T_2 to the output of T_1 is linear, and it exists, so there exists $S \in \mathcal{L}(W)$ such that $T_1 = ST_2$.

5 If $\text{range } T_1 = \text{range } T_2$, then there exists an invertible operator that's linear and maps from V to V such that $T_1 = T_2S$.

6

9 Observe that if ST is invertible, then we must be able to multiply it by any S^{-1} or T^{-1} , so that we could find S or T . And if S and T are invertible, their determinants are nonzero, so ST 's determinant is also nonzero, so it's also invertible.

10 If $ST = I$, then ST is invertible, so $(ST)^{-1} = T^{-1}S^{-1} = I$, so multiplying by T on the left gives $S^{-1} = T$, so multiplying by S on the right gives $TS = I$.

11 If $STU = I$, multiplying by S^{-1} on the left gives $TU = S^{-1}$, then multiplying by U^{-1} on the right gives $T = S^{-1}U^{-1}$, which we know equals $(US)^{-1}$ by problem 1, so $T^{-1} = US$, as desired.

12 If V isn't finite-dimensional, then S , T , and U might not necessarily have inverses, so there goes the proof.

13 If R , S , and T are all linear maps from V to V itself, since RST is plugging the output of T into S and plugging this into R , then since T is surjective, then S is injective.

18 It's a linear map from $\mathbb{F}$ to V . Since $\dim \mathbb{F} = 1$, then it's isomorphic.

19

(a) Since $\deg Tp \leq \deg p$ for all p , this must imply that T is a bijection (since it's also injective), so it must be surjective.

(b) Since its surjective, then $\deg Tp = \deg p$ otherwise it wouldn't be surjective. Tada.

20 Observe that if the only solution if $c_1 = \cdots = c_n = 0$ is all 0's, then this means that solutions exist, and its linearly independent ($A_{\cdot,k}$ for all k) so it can also act as a basis, so there will have to exist x_k . And if there exist x_k as solutions to the c , then it's a basis, so it's linearly independent, so $x_1 = \cdots = x_n = 0$ is the only solution if all the c 's equal 0.

3.E Products and Quotients of Vector Spaces

1 Consider the map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^2$. Then if we map $T(a, b, c, d, e)$ to $(a, b, 0, 0, 0)$, then the range has dimension 2, and the null space consists of all quintuplets of the form $(0, 0, c, d, e)$, which has dimension 3.

2 Suppose $S : X \rightarrow Y$ and $T : U \rightarrow V$. Then if we have $x \in X$, which maps to a $y \in Y$, then $y \in \text{null } T$, so $T(y) = T(S(x)) = 0$, so $TS = 0$, so $(ST)^2 = 0$.

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we let U be the set consisting of all inputs that don't result in an output of 0, plus one element from the null set, which is $\{0\}$.

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14 Since $\text{null } T = U$, then since U is 3-dimensional, then $\dim \text{null } T = 3$ as well. $\dim \text{null } T + \dim \text{range } T = \dim V = 8$, so $\dim \text{range } T = 5 = \dim W$, so T is surjective, as desired.

15 So the null space consists of quintuplets of the form $(3a, a, b, b, b)$, which has dimension 2. Since $\dim \text{range } T = \dim V - \dim \text{null } T = 5 - 2 = 3$, then $\dim \text{range } T > \dim W$, which is impossible.

16 Thus there exists values such that

$$Tv = \varphi_1(v)w_1 + \cdots + \varphi_m(v)w_m.$$

17 Consider the map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^2$. Then if we map $T(a, b, c, d, e)$ to $(a, b, 0, 0, 0)$, then the range has dimension 2, and the null space consists of all quintuplets of the form $(0, 0, c, d, e)$, which has dimension 3.

18 Suppose $S : X \rightarrow Y$ and $T : U \rightarrow V$. Then if we have $x \in X$, which maps to a $y \in Y$, then $y \in \text{null } T$, so $T(y) = T(S(x)) = 0$, so $TS = 0$, so $(ST)^2 = 0$.

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3.F Duality

1 Observe that if both T (which is a linear map from U to V) and S which is a linear map from V to W are invertible, then ST , which is a linear map from U to W is invertible, since it's the product of S and T , and thus $ST = T^{-1}S^{-1}$.

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3 If there exists an invertible operator $T \in \mathcal{L}(V)$ such that $Tu = Su$ for every $u \in U$, then this must mean that since T is both injective and surjective, then since $Tu = Su$, then so must S . Conversely, if S is injective, then there must exist T such that $Tu = Su$.

4 If $\text{null } T_1 = \text{null } T_2$, this implies that the map from the output of T_2 to the output of T_1 is linear, and it exists, so there exists $S \in \mathcal{L}(W)$ such that $T_1 = ST_2$.

5 If $\text{range } T_1 = \text{range } T_2$, then there exists an invertible operator that's linear and maps from V to V such that $T_1 = T_2S$.

6 Since T is a linear map from V to W and $v_1, \dots, v_n$ spans V this also means that $Tv_1, \dots, Tv_n$ spans W .

7 $(S_1S_2 \cdots S_n)(v) = (S_1(v))(S_2(v)) \cdots (S_n(v))$. So if we have another w such that $(S_1S_2 \cdots S_n)(w) = (S_1S_2 \cdots S_n)(v)$, then we must have $S_1(v) = S_1(w)$, etc, so since $S_1, \dots, S_n$ are injective, then $v = w$.

8 We want to show that T is surjective. Obviously, if T is surjective, then U suffices as the input because we really only need one zero, and the rest would be taken care of if we let U be the set consisting of all inputs that don't result in an output of 0, plus one element from the null set, which is $\{0\}$.

9 Observe that if ST is invertible, then we must be able multiply it by any S^{-1} or T^{-1} , so that we could find S or T . And if S and T are invertible, their determinants are nonzero, so ST 's determinant is also nonzero, so its also invertible.

10 If $ST = I$, then ST is invertible, so $(ST)^{-1} = T^{-1}S^{-1} = I$, so multiplying by T on the left gives $S^{-1} = T$, so multiplying by S on the right gives $TS = I$.

11 If $STU = I$, multiplying by S^{-1} on the left gives $TU = S^{-1}$, then multiplying by U^{-1} on the right gives $T = S^{-1}U^{-1}$, which we know equals $(US)^{-1}$ by problem 1, so $T^{-1} = US$, as desired.

12 describe the linear functionals $T'(\varphi_1)$ and $T'(\varphi_2)$. if we write $T'(\varphi)$ and put it in, so we get something along the lines f

13

(a) Let φ_1 be the functional that maps $\begin{pmatrix} a \\ b \end{pmatrix}$ to a and φ_2 be the functional that maps $\begin{pmatrix} a \\ b \end{pmatrix}$ to b . Since T' represents the linear map from $(\mathbb{R}^2)'$ to $(\mathbb{R}^3)'$, then $T'(\varphi_1)$ and $T'(\varphi_2)$ are linear maps from $\mathbb{R}^2$ to $\mathbb{R}$.

(b) ψ_1 would be the functional that maps $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ to a , ψ_2 would be the functional that maps $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ to b , and ψ_3 would be the functional that maps $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ to c . Then, since ψ_1, ψ_2 , and ψ_3 are parts of a basis, so they are (basis)es though not really, so $T'(\varphi_1)$ and $T'(\varphi_2)$ are linear combinations of ψ_1, ψ_2, ψ_3 .

14

(a) Well, T is a linear map, and $(Tp)(x)$ is a linear transformation on $p(x)$. Then since $\varphi(p)$ is a functional that maps $\mathcal{P}(\mathbb{R})$ to $\mathbb{R}$. T is a linear map from $\mathcal{P}(\mathbb{R})$ to $\mathcal{P}(\mathbb{R})$, so T' is a linear map from $\mathcal{P}(\mathbb{R})'$ to $\mathcal{P}(\mathbb{R})'$, which is basically a linear map from (the set of functionals from $\mathcal{P}(\mathbb{R})$ to $\mathbb{R}$) to itself. Thus $T'(\varphi) = \varphi(Tp) = \varphi(x^2p(x) + p''(x))$. Since $(x^2p(x) + p''(x))' = 2xp(x) + x^2p'(x) + p'''(x)$, then plugging in $x = 4$ gives $8p(4) + 16p'(4) + p'''(4)$.

(b) Plugging x^3 into T gives $(Tp)(x^3) = x^6p(x^3) + p''(x^3)$. Then $\varphi(Tp(x^3)) = \varphi(x^6p(x^3) + p''(x^3))$. First, taking the derivative and applying the chain rule gives

$$[x^6p(x^3) + p''(x^3)]' = 6x^5p(x^3) + x^6 \cdot 3x^2p'(x^3) + 3x^2p'''(x^3).$$

Then we plug in $x = 4$ to get an gigantic expression.

15 Well, since T' maps from $\mathcal{L}(W, \mathbb{R})$ to $\mathcal{L}(V, \mathbb{R})$, then if $T' = 0$, it would also imply that T , which maps from V to W equals 0, and vice versa.

16 Since T is an element of $\mathcal{L}(V, W)$ and T' is an element of $\mathcal{L}(W', V')$, and since these two are distinct and don't have any common elements, it suffices to prove that they have an equal number of elements. Since $(v_1, \dots, v_m)$ encompasses all of V , then the map from the vectors, $\mathbb{F}^m$, to V must be surjective. If $v_1, \dots, v_m$ are linearly independent, this means that the null space has degree 0, and consists of only $\{0\}$.

17 Obviously, if $\varphi(u) = 0$ for some u , then u is automatically an element of null φ . Therefore, if for all $u \in U$, $\varphi(u) = 0 \implies u \in \text{null } \varphi$, then $U \subset \text{null } \varphi$.

18 If $U = \{0\}$, then for the only $u \in U$ is $u = 0$. Obviously, for all $\varphi \in V'$, $\varphi(u) = 0$ for any $u \in U$, namely because there's only one value of u and $\varphi(0) = 0$. Therefore, if

$U = \{0\}$, then $U^0 = V'$. Conversely, if for all $\varphi \in V'$, $\varphi(u) = 0$ for every $u \in U$, wouldn't this imply that V' is severely limited? So U can only have one value, namely 0.

19 Observe that if $U^0 = 0$, then only the function $\varphi = 0$ satisfies $\varphi(u) = 0$ for all $u \in U$, so $U = V$. And if $U = V$, then for $\varphi(u) = 0$, we can only have $\varphi = 0$, because then the functionals in V would be constants, so they'd have to equal 0.

20 Consider U^0 , which consists of all $\varphi \in V'$ such that for all $u \in U$, $\varphi(u) = 0$. And we can also consider W^0 , which consists of all $\varphi \in V'$ such that for all $w \in W$, $\varphi(w) = 0$. Note that if $u \in U$, then $u \in W$, so if $\varphi \in W^0$, then we also have $\varphi \in U^0$, so $W^0 \subset U^0$.

21 If $W^0 \subset U^0$, then this implies that for any $\varphi \in V'$, W^0 consists of the φ such that $\varphi(w) = 0$ for all $w \in W$, and U^0 consists of the φ such that $\varphi(u) = 0$ for all $u \in U$. Since $W^0 \subset U^0$, then if we have any φ such that $\varphi(w) = 0$ for all $w \in W$, then $\varphi(u) = 0$ for all $u \in U$. Thus $U \subset W$, as desired.

22 Consider any $x \in U + W$. Automatically, $x \in U$ and $x \in W$. Therefore, if for any $\varphi \in V'$ such that $\varphi \in (U + W)^0$, then for any $x \in U + W \iff x \in U, x \in W$, $\varphi(x) = 0$. Thus, if $\varphi(x) = 0$ for all x , then for the same φ , we also have $\varphi(u) = 0$ for all $u \in U$ and $\varphi(w) = 0$ for all $w \in W$. Hence, $(U + W)^0 \subset U^0 \cap W^0$.

However, $U^0 \cap W^0$ consists of all $\varphi \in V'$ such that $\varphi(u) = 0$ for all $u \in U$ and $\varphi(w) = 0$ for all $w \in W$. If, for any x , $x \in U$ and $x \in W$, then $x \in U + W$, so this also implies that $\varphi(x) = 0$ for all $x \in U + W$, so $U^0 \cap W^0 \subset (U + W)^0$, so $U^0 \cap W^0 = (U + W)^0$, as desired.

23 If $x \in U \cap W$, then $x \in U$ and $x \in W$. Therefore, if $\varphi(x) = 0$ for all $x \in U \cap W$, then $\varphi(x) = 0$ for all $x \in U$, and $\varphi(x) = 0$ for all $x \in W$. Hence, $(U \cap W)^0 \subset U^0 + W^0$.

Additionally, if $\varphi \in U^0 + W^0$, then $\varphi(u) = 0$ for all $u \in U$ and $\varphi(w) = 0$ for all $w \in W$, so $\varphi(x) = 0$ if $x \in U \cap W$. Thus $(U \cap W)^0 = U^0 + W^0$.

24 Consider a basis $u_1, \dots, u_m$ of U , and extend it to a basis $u_1, \dots, u_m, \dots, u_n$ of V , so basically $\dim U = m$ and $\dim V = n$. Now let $\varphi_1, \dots, \varphi_m, \dots, \varphi_n$ be the dual basis of V' , so then $\varphi_{m+1}, \dots, \varphi_n$ can be a basis of U^0 . Thus $\dim U^0 = n - m$, so $\dim U + \dim U^0 = \dim V$, as desired.

25 If $\varphi(v) = 0$ for every $\varphi \in U^0$ and we are looking for the v , then since $\varphi \in U^0$, we already know that $\varphi(v) = 0$ for all $v \in V$. Thus the set is U .

26 This is like taking the reverse of the reverse; it goes back to the original. We're looking for the v , then taking the annihilator or something. So this gives $\Gamma = \{v \in V : \varphi(v) = 0 \text{ for every } \varphi \in \Gamma\}^0$.

27 T' is the linear map from $\mathcal{L}(\mathcal{P}_5(\mathbb{R}), \mathbb{R})$ to $\mathcal{L}(\mathcal{P}_5(\mathbb{R}), \mathbb{R})$. Thus, the range of T consists of all p such that $\varphi(p) = 0$. But $\varphi(p) = p(8)$, so it consists of all p such that $p(8) = 0$.

28 Observe that if both T (which is a linear map from U to V) and S which is a linear map from V to W are invertible, then ST , which is a linear map from U to W is invertible, since it's the product of S and T , and thus $ST = T^{-1}S^{-1}$.

29 This doesn't hold because the sum of two noninvertible operators might just be invertible.

30 If there exists an invertible operator $T \in \mathcal{L}(V)$ such that $Tu = Su$ for every $u \in U$, then this must mean that since T is both injective and surjective, then since $Tu = Su$, then so must S . Conversely, if S is injective, then there must exist T such that $Tu = Su$.

31 If V is finite-dimensional and $\varphi_1, \dots, \varphi_n$ is a basis of V' . Then there must be a basis of V whose dual basis is $\varphi_1, \dots, \varphi_n$ because V' is the dual space of V . V' is the set of all linear maps from V to $\mathbb{F}$, and since $\varphi_1, \dots, \varphi_n$ are the functionals such that there exists a basis $v_1, \dots, v_n$ of V such that $\varphi_i(v_j) = 1$ if and only if $i = j$, and 0 otherwise.

32 We show that everything is equivalent to T is invertible. Obviously, if the columns of $\mathcal{M}(T)$ are linearly independent in $\mathbb{F}^{n,1}$, then it can form a basis (since its linearly independent, and everything that's linearly independent can form a basis), so they are

equivalent. Then, if they span $\mathbb{F}^{n,1}$, T is invertible. Similarly, if the rows of $\mathcal{M}(T)$ are linearly independent, it's equivalent to being a basis of $\mathbb{F}^{1,n}$, so it spans $\mathbb{F}^{1,n}$, and vice-versa. Therefore, these 5 statements are equivalent to each other.

33. If $U = \{0\}$, then for the only $u \in U$ is $u = 0$. Obviously, for all $\varphi \in V'$, $\varphi(u) = 0$ for any $u \in U$, namely because there's only one value of u and $\varphi(0) = 0$. Therefore, if $U = \{0\}$, then $U^0 = V'$. Conversely, if for all $\varphi \in V'$, $\varphi(u) = 0$ for every $u \in U$, wouldn't this imply that V' is severely limited? So U can only have one value, namely 0.

34

- (a) We already know that Λ maps from V to V'' , so it suffices to show that this transformation is linear. Since $(\Lambda v)(\varphi) = \varphi(v)$, if we have φ_1 and φ_2 , then

$$(\Lambda v)(\varphi_1 + \varphi_2) = (\varphi_1 + \varphi_2)(v) = \varphi_1(v) + \varphi_2(v) = (\Lambda v)(\varphi_1) + (\Lambda v)(\varphi_2).$$

Furthermore,

$$(\Lambda v)(\lambda\varphi) = (\lambda\varphi)(v) = \lambda \cdot \varphi(v) = \lambda(\Lambda v)(\varphi).$$

Therefore, Λ is a linear map from V to V'' .

- (b) . Consider U^0 , which consists of all $\varphi \in V'$ such that for all $u \in U$, $\varphi(u) = 0$. And we can also consider W^0 , which consists of all $\varphi \in V'$ such that for all $w \in W$, $\varphi(w) = 0$. Note that if $u \in U$, then $u \in W$, so if $\varphi \in W^0$, then we also have $\varphi \in U^0$, so $W^0 \subset U^0$.
- (c) We already know that Λ is a linear map from V onto V'' , so we wish to prove that if V is finite-dimensional, $\dim V = \dim V''$. V'' is the set of linear maps from V' to $\mathbb{R}$, so its the set of linear maps from $\mathcal{L}(V, \mathbb{R})$ to $\mathbb{R}$. Thus, we have an isomorphism.

35 $(\mathcal{P}(\mathbb{R}))'$ is the set of linear maps from $\mathcal{P}(\mathbb{R})$ to $\mathbb{R}$. And $\mathbb{R}^\infty$ is the set of all sequences over $\mathbb{R}$. Since a polynomial is basically a sequence of real numbers (take the coefficients), then via a one-to-one correspondence, we're done.

36

- (a) U^0 consists of all $\varphi \in V'$ such that $\varphi(u) = 0$ for all $u \in U$. However, we know that $i(u) = u$ for all $u \in U$. Since i' is a map from V' to U' , then it's a map from φ to U' . The null space of i' is the set of i such that $i(u) = 0$ for all $u \in U$, which is equivalent to U^0 .
- (b) i' is literally a linear map from V' to U' , so of course its range is U' .
- (c) We already know that i is a linear map from U to V , so i' is a linear map from V' to U' , so $\tilde{i}'$ is the linear map from $V'/(i') \rightarrow U'$. But, $i' = U^0$ by part (a), so its the linear map from V'/U^0 to U' , as desired.

37

- (a) It suffices to show that the null space of π' , which is the set of linear transformations $\in \mathcal{L}(V/U, \mathbb{R})$ that map to 0, has only one element, which is 0. Since V/U is the set of all affine subsets of V parallel to U , then we are done.
- (b) Since U^0 is defined as the set of $\varphi \in V'$ such that $\varphi(u) = 0$ for all $u \in U$, then since π' maps to functionals in V' , and $\varphi \in V'$ as well, then $\text{range } \pi' = U^0$, as desired.
- (c) Since $\text{range } \pi' = U^0$, then π' maps from its domain, $(V/U)'$, to U^0 , and thus its a linear map from $(V/U)'$ to U^0 . And, since its injective, and $\dim(V/U)' = \dim U^0$, so its also surjective. Since its both injective and surjective, then its bijective, so its an isomorphism from $(V/U)'$ onto U^0 .

Chapter 4: Polynomials

Linear Algebra Done Right

Carissa Zeng

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1

sum of z and $\bar{z}$: Let $z = a + bi$, so $\bar{z} = a - bi$, so $z + \bar{z} = 2a = 2\operatorname{Re} z$.

difference of z and $\bar{z}$: Let $z = a + bi$, so $\bar{z} = a - bi$, so $z - \bar{z} = 2bi = 2(\operatorname{Im} z)i$.

product of z and $\bar{z}$: $z\bar{z} = (a + bi)(a - bi) = a^2 + b^2 = (\sqrt{a^2 + b^2})^2 = |z|^2$.

additivity and multiplicativity of complex conjugate: Let $w = a + bi$ and $z = c + di$. $w + z = (a + c) + (b + d)i$, so $\overline{w + z} = (a + c) - (b + d)i$. $\bar{w} = a - bi$ and $\bar{z} = c - di$, so $\bar{w} + \bar{z} = (a + c) - (b + d)i = \overline{w + z}$. $wz = (a + bi)(c + di) = (ac - bd) + (bc + da)i$, so $\overline{wz} = (ac - bd) - (bc + da)i$. $\bar{w} \cdot \bar{z} = (a - bi)(c - di) = (ac - bd) - (bc + da)i = \overline{wz}$, as desired.

conjugate of conjugate: $z = a + bi$, $\bar{z} = a - bi$, $\bar{\bar{z}} = \overline{a - bi} = a + bi = z$, as desired.

real and imaginary parts are bounded by $|z|$: Suppose $z = a + bi$, so that $\operatorname{Re} z = a$ and $\operatorname{Im} z = b$. Both a and b are real numbers. Then $|z| = \sqrt{a^2 + b^2}$, so $|z|^2 = a^2 + b^2$. Since they are real numbers, $b^2 \geq 0$ and $a^2 \geq 0$, so $a^2 \leq |z|^2 \implies |a| \leq |z|$ and $b^2 \leq |z|^2 \implies |b| \leq |z|$.

absolute value of the complex conjugate: Let $z = a + bi$, so $\bar{z} = a - bi$. $|z| = \sqrt{a^2 + b^2}$ and $|\bar{z}| = \sqrt{a^2 + (-b)^2} = \sqrt{a^2 + b^2} = |z|$, as desired.

multiplicativity of absolute value: Let $w = a + bi$, $z = c + di$. Then $wz = (a + bi)(c + di) = ac - bd + (bc + da)i$, and

$$\begin{aligned} |wz|^2 &= (ac - bd)^2 + (bc + da)^2 \\ &= (a^2c^2 - 2abcd + b^2d^2) + (b^2c^2 + 2abcd + d^2a^2) \\ &= a^2c^2 + b^2d^2 + b^2c^2 + d^2a^2 \end{aligned}$$

$$|w|^2|z|^2 = (a^2 + b^2)(c^2 + d^2) = a^2c^2 + b^2c^2 + b^2d^2 + a^2d^2 = |wz|^2,$$

so since $|w|^2|z|^2 = |wz|^2$, then $|w||z| = |wz|$.

2 No, not necessarily. For example consider, say, $m = 2$, and the polynomials $p_1(x) = x^2$, $p_2(x) = -x^2$. Then $p_1(x) + p_2(x) = 0$, which has degree 0, so its not closed under addition, so its not a subspace of $\mathcal{P}(\mathbb{F})$.

3 No, not necessarily. For example, consider the polynomials $p_1(x) = x^2 + x$ and $p_2(x) = -x^2 + x$. Then $p_1(x) + p_2(x) = 2x$, which is an odd degreed polynomial, so its not closed under addition, so its not a subspace of $\mathcal{P}(\mathbb{F})$.

4 If $m \leq n$, we can go with something like $p = (x - \lambda_1)^{n-m+1}(x - \lambda_2) \cdots (x - \lambda_m)$, which certainly has degree n and has only $\lambda_1, \dots, \lambda_m$ as zeros.

5 We can use the method of undetermined coefficients, but if we let $z_1, \dots, z_{m+1}$ be a basis of $\mathbb{F}$, then we can also let $w_1, \dots, w_{m+1}$ be its dual basis, so there does exist p such that $p(z_j) = w_j$.

6 If p has m distinct zeros, then this means that each zero occurs exactly once, so p looks something like $p(x) = c(x - \lambda_1) \cdots (x - \lambda_m)$. Then $p'(\lambda_j) \neq 0$ for any $1 \leq j \leq m$, so p and p' have no zeros in common, and the reverse is also true.

7 Well, if you need it to have real coefficients, then the imaginary roots have to come in pairs, so there are an even number of imaginary roots. However, if we have a polynomial of odd degree, then there must be an odd number of roots, so there has to be at least one root that is real.

8 Observe that if p is a polynomial, then p' is also a polynomial, and when $x = 3$, TP is a constant. Now suppose $x \neq 3$. Let $p(x) = k + (x - 3)q(x)$ where q is also a polynomial, so k is the remainder when $p(x)$ is divided by $x - 3$. Then $p(3) = k$, so $p(x) - k = p(x) - p(3) = (x - 3)q(x)$. Hence, $\frac{p(x) - p(3)}{x - 3} = q(x)$, which is also a polynomial. Thus $TP \in \mathcal{P}(\mathbb{R})$.

Also, observe that T is continuous because $\lim_{x \rightarrow 3} \frac{p(x) - p(3)}{x - 3} = p'(3)$ by the definition of derivative, so T is a linear map.

9 Observe that if we let $p(z) = c_0 + c_1z + \cdots + c_nz^n$, then

$$\begin{aligned} p(\bar{z}) &= c_0 + c_1\bar{z} + c_2\bar{z}^2 + \cdots + c_n\bar{z}^n \\ &= c_0 + c_1\bar{z} + c_2\overline{z^2} + \cdots + c_n\overline{z^n} \\ &= \overline{c_0 + c_1z + c_2z^2 + \cdots + c_nz^n} \\ &= \overline{p(z)}. \end{aligned}$$

Thus $\overline{p(\bar{z})} = p(z)$, so $q(z) = p(z)^2$, so it has real coefficients.

10 We know that if we can determine the value of p with degree m for $m + 1$ values of x , then we can determine p . Let $p = c_0 + c_1x + \cdots + c_mx^m$, so we know that

$$\begin{aligned} p(x_0) &= c_0 + c_1x_0 + \cdots + c_mx_0^m \\ p(x_1) &= c_0 + c_1x_1 + \cdots + c_mx_1^m \\ &\vdots \\ p(x_m) &= c_0 + c_1x_m + \cdots + c_mx_m^m. \end{aligned}$$

Since we have $m + 1$ unknowns, which are $c_0, \dots, c_m$, and $m + 1$ equations, then we can solve for the c_i . Since all of the x_i are real numbers, and all of the $p(x_i)$ are real numbers, then all of the c_i are also real numbers.

11

(a) U is the set of all polynomials, times p . Thus, $\dim \mathcal{P}(\mathbb{F})/U = \dim(p + p\mathcal{P}) = \dim p$.

(b) We can use $1, x, \dots, x^{\deg p - 1}$.

Chapter 5: Eigenvalues, Eigenvectors, and Invariant Subspaces

Linear Algebra Done Right

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5.A Invariant Subspaces

1

(a) No, not necessarily. For example consider, say, $m = 2$, and the polynomials $p_1(x) = x^2$, $p_2(x) = -x^2$. Then $p_1(x) + p_2(x) = 0$, which has degree 0, so its not closed under addition, so its not a subspace of $\mathcal{P}(\mathbb{F})$.

(b) No, not necessarily. For example, consider the polynomials $p_1(x) = x^2 + x$ and $p_2(x) = -x^2 + x$. Then $p_1(x) + p_2(x) = 2x$, which is an odd degreed polynomial, so its not closed under addition, so its not a subspace of $\mathcal{P}(\mathbb{F})$.

2 If $m \leq n$, we can go with something like $p = (x - \lambda_1)^{n-m+1}(x - \lambda_2) \cdots (x - \lambda_m)$, which certainly has degree n and has only $\lambda_1, \dots, \lambda_m$ as zeros.

3 We can use the method of undetermined coefficients, but if we let $z_1, \dots, z_{m+1}$ be a basis of $\mathbb{F}$, then we can also let $w_1, \dots, w_{m+1}$ be its dual basis, so there does exist p such that $p(z_j) = w_j$.

4 If p has m distinct zeros, then this means that each zero occurs exactly once, so p looks something like $p(x) = c(x - \lambda_1) \cdots (x - \lambda_m)$. Then $p'(\lambda_j) \neq 0$ for any $1 \leq j \leq m$, so p and p' have no zeros in common, and the reverse is also true.

5 Well, if you need it to have real coefficients, then the imaginary roots have to come in pairs, so there are an even number of imaginary roots. However, if we have a polynomial of odd degree, then there must be an odd number of roots, so there has to be at least one root that is real.

6 Observe that if p is a polynomial, then p' is also a polynomial, and when $x = 3$, Tp is a constant. Now suppose $x \neq 3$. Let $p(x) = k + (x - 3)q(x)$ where q is also a polynomial, so k is the remainder when $p(x)$ is divided by $x - 3$. Then $p(3) = k$, so $p(x) - k = p(x) - p(3) = (x - 3)q(x)$. Hence, $\frac{p(x) - p(3)}{x - 3} = q(x)$, which is also a polynomial. Thus $Tp \in \mathcal{P}(\mathbb{R})$.

Also, observe that T is continuous because $\lim_{x \rightarrow 3} \frac{p(x) - p(3)}{x - 3} = p'(3)$ by the definition of derivative, so T is a linear map.

7 If $T(x, y) = \lambda(-3y, x)$, then $x = -3\lambda y$ and $y = \lambda x$, so $x = -3\lambda^2 x$, so if λ exists, we must have $x = 0$. Then $y = \lambda x = 0$, which is a contradiction, since we can't have $x = y = 0$, so λ doesn't exist.

8 If we $T(w, z) = \lambda(z, w)$, then $w = \lambda z$ and $z = \lambda w = \lambda^2 z$, so $\lambda = \pm 1$. If $\lambda = 1$, then the set of all (a, a) with $a \in \mathbb{R}$. If $\lambda = -1$, the set of all $(a, -a)$ with $a \in \mathbb{R}$. Also $a \neq 0$.

9 $T(z_1, z_2, z_3) = (2z_2, 0, 5z_3) = \lambda(z_1, z_2, z_3)$. $2z_2 = \lambda z_1$, $0 = \lambda z_2$, and $5z_3 = \lambda z_3$. From the last equation, either $\lambda = 5$ or $z_3 = 0$. If $\lambda = 5$, then $z_2 = 0$, and $z_1 = 0$, so we would have $\lambda = 5$ and all vectors of the form $(0, 0, a)$ with $a \in \mathbb{R}$ and $a \neq 0$. If $z_3 = 0$, then we have $z_2 = 0$ or $\lambda = 0$. If $z_2 = 0$, then $z_1 = 0$, so nope. If $\lambda = 0$, then $z_2 = 0$, so nope as well.

10

- (a) The eigenvalues are $1, \dots, n$. If $\lambda = i$, then let $x_1 = \dots = x_{i-1} = x_{i+1} = \dots = x_n = 0$ and $x_i = \text{anything}$.
- (b) The invariant subspaces are the set of integers, rational numbers, irrational numbers, etc.

11 $p' = \lambda p$ for some p , so we have $\lambda = 0$ and p is a constant. There's no other way because $(x^n)' = nx^{n-1}$, and it's impossible for nx^{n-1} to equal λx^n .

12 $(Tp)(x) = xp'(x)$, so if we have x^n , then $x(x^n)' = nx^n$. Thus, if we let $n \in \mathbb{N}$ and $\lambda = n$, then any $p(x) = cx^n$ where c is a constant will do.

13 Observe that if we let $p(z) = c_0 + c_1z + \dots + c_nz^n$, then

$$\begin{aligned} p(\bar{z}) &= c_0 + c_1\bar{z} + c_2\bar{z}^2 + \dots + c_n\bar{z}^n \\ &= c_0 + c_1\bar{z} + c_2\overline{z^2} + \dots + c_n\overline{z^n} \\ &= \overline{c_0 + c_1z + c_2z^2 + \dots + c_nz^n} \\ &= \overline{p(z)}. \end{aligned}$$

Thus $\overline{p(\bar{z})} = p(z)$, so $q(z) = p(z)^2$, so it has real coefficients.

14 We know that if we can determine the value of p with degree m for $m+1$ values of x , then we can determine p . Let $p = c_0 + c_1x + \dots + c_mx^m$, so we know that

$$\begin{aligned} p(x_0) &= c_0 + c_1x_0 + \dots + c_mx_0^m \\ p(x_1) &= c_0 + c_1x_1 + \dots + c_mx_1^m \\ &\vdots \\ p(x_m) &= c_0 + c_1x_m + \dots + c_mx_m^m. \end{aligned}$$

Since we have $m+1$ unknowns, which are $c_0, \dots, c_m$, and $m+1$ equations, then we can solve for the c_i . Since all of the x_i are real numbers, and all of the $p(x_i)$ are real numbers, then all of the c_i are also real numbers.

15

- (a) Let $z = a + bi$, so $\bar{z} = a - bi$, so $z + \bar{z} = 2a = 2\operatorname{Re} z$. Let $z = a + bi$, so $\bar{z} = a - bi$, so $z - \bar{z} = 2bi = 2(\operatorname{Im} z)i$. $z\bar{z} = (a + bi)(a - bi) = a^2 + b^2 = (\sqrt{a^2 + b^2})^2 = |z|^2$.
- (b) Let $w = a + bi$ and $z = c + di$. $w + z = (a + c) + (b + d)i$, so $\overline{w + z} = (a + c) - (b + d)i$. $\bar{w} = a - bi$ and $\bar{z} = c - di$, so $\bar{w} + \bar{z} = (a + c) - (b + d)i = \overline{w + z}$. $wz = (a + bi)(c + di) = (ac - bd) + (bc + da)i$, so $\overline{wz} = (ac - bd) - (bc + da)i$. $\bar{w} \cdot \bar{z} = (a - bi)(c - di) = (ac - bd) - (bc + da)i = \overline{wz}$, as desired.

16 If $Tv = \lambda v$ for some v , then since T is a linear transformation from V to V , λv is also an element of V , so v is an element of V in the first place, so $\bar{\lambda}v$ is also an element of V , so $\bar{\lambda}$ has an eigenvalue, namely $\bar{v}$.

17 Consider the map $(x_1, x_2, x_3, x_4) \rightarrow (0, x_1x_4, x_2, x_3)$. We would need $x_1 = 0$, so $x_2 = 0$, so $x_3 = 0$, so $x_4 = 0$.

18 We would need $z_1 = \lambda \cdot 0 = 0$, $z_2 = \lambda \cdot z_1 = 0$, $z_3 = \lambda \cdot z_2 = 0$, etc. But we only have an eigenvalue when the input is nonzero, so there would be no eigenvalue.

19 Since the entries of the output all equal 0, then since

$$T(x_1, \dots, x_n) = \lambda(x_1, \dots, x_n) = (x_1 + \dots + x_n, \dots, x_1 + \dots + x_n),$$

then $x_1 = \dots = x_n$, so as long as we have that, we can let $\lambda = n$, and the eigenvectors are of the form $(a, a, \dots, a)$, where $a \in \mathbb{R}$, and $a \neq 0$.

20 We need $z_1 = \lambda z_2$, $z_2 = \lambda z_3$, etc, so $z_1 = \lambda z_2 = \lambda^2 z_3 = \dots$. Therefore, the answer is clear. We have λ and $(a, \frac{a}{\lambda}, \frac{a}{\lambda^2}, \dots)$ as the eigenvector.

21

- (a) If $Tv = \lambda v$ for some vector v , then $T^{-1}v = \frac{1}{\lambda}v$, so $\frac{1}{\lambda}$ is an eigenvalue if λ is.
- (b) Since λ to T and $\frac{1}{\lambda}$ to T^{-1} are eigenvalues to the same eigenvectors, then since λ being an eigenvalue implies that $\frac{1}{\lambda}$ is an eigenvalue, then T and T^{-1} have the same eigenvectors.

22 $T(Tv) = T(3w) = 3Tw = 9v$ since T is a linear map and $Tv = 3w$ and $Tw = 3v$, so 9 is an eigenvalue of T^2 . Thus, 3 or -3 is an eigenvalue of T .

23 Suppose $z = a + bi$, so that $\operatorname{Re} z = a$ and $\operatorname{Im} z = b$. Both a and b are real numbers. Then $|z| = \sqrt{a^2 + b^2}$, so $|z|^2 = a^2 + b^2$. Since they are real numbers, $b^2 \geq 0$ and $a^2 \geq 0$, so $a^2 \leq |z|^2 \implies |a| \leq |z|$ and $b^2 \leq |z|^2 \implies |b| \leq |z|$.

24

1. Let $w = a + bi$, $z = c + di$. Then $wz = (a + bi)(c + di) = ac - bd + (bc + da)i$, and

$$\begin{aligned} |wz|^2 &= (ac - bd)^2 + (bc + da)^2 \\ &= (a^2c^2 - 2abcd + b^2d^2) + (b^2c^2 + 2abcd + d^2a^2) \\ &= a^2c^2 + b^2d^2 + b^2c^2 + d^2a^2 \end{aligned}$$

2.

$$|w|^2|z|^2 = (a^2 + b^2)(c^2 + d^2) = a^2c^2 + b^2c^2 + b^2d^2 + a^2d^2 = |wz|^2,$$

so since $|w|^2|z|^2 = |wz|^2$, then $|w||z| = |wz|$.

25 Let $z = a + bi$, so $\bar{z} = a - bi$. $|z| = \sqrt{a^2 + b^2}$ and $|\bar{z}| = \sqrt{a^2 + (-b)^2} = \sqrt{a^2 + b^2} = |z|$, as desired.

26 Observe that the null space consists of points of the form $(5a, a, 7b, b)$, which has dimension 2. Since $\dim \operatorname{null} T + \dim \operatorname{range} T = \dim V = 4$, then $\dim \operatorname{range} T = 2 = \dim W$, so T is surjective, as desired.

Since $\operatorname{null} T = U$, then since U is 3-dimensional, then $\dim \operatorname{null} T = 3$ as well. $\dim \operatorname{null} T + \dim \operatorname{range} T = \dim V = 8$, so $\dim \operatorname{range} T = 5 = \dim W$, so T is surjective, as desired.

27 So the null space consists of quintuplets of the form $(3a, a, b, b, b)$, which has dimension 2. Since $\dim \operatorname{range} T = \dim V - \dim \operatorname{null} T = 5 - 2 = 3$, then $\dim \operatorname{range} T > \dim W$, which is impossible. Thus there exists values such that

$$Tv = \varphi_1(v)w_1 + \dots + \varphi_m(v)w_m.$$

28 Consider the map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^2$. Then if we map $T(a, b, c, d, e)$ to $(a, b, 0, 0, 0)$, then the range has dimension 2, and the null space consists of all quintuplets of the form $(0, 0, c, d, e)$, which has dimension 3. Suppose $S : X \rightarrow Y$ and $T : U \rightarrow V$. Then if we

have $x \in X$, which maps to a $y \in Y$, then $y \in \text{null } T$, so $T(y) = T(S(x)) = 0$, so $TS = 0$, so $(ST)^2 = 0$.

29 Since $(v_1, \dots, v_m)$ encompasses all of V , then the map from the vectors, $\mathbb{F}^m$, to V must be surjective. If $v_1, \dots, v_m$ are linearly independent, this means that the null space has degree 0, and consists of only $\{0\}$. Since we know that $\dim \text{null } T + \dim \text{range } T = \dim V$ but $\dim V = 5$ and $\dim \text{null } T > 2$ and $\dim \text{range } T = 4$, which is impossible.

30 Observe that if both T (which is a linear map from U to V) and S which is a linear map from V to W are invertible, then ST , which is a linear map from U to W is invertible, since it's the product of S and T , and thus $ST = T^{-1}S^{-1}$. This doesn't hold because the sum of two noninvertible operators might just be invertible.

31 If there exists an invertible operator $T \in \mathcal{L}(V)$ such that $Tu = Su$ for every $u \in U$, then this must mean that since T is both injective and surjective, then since $Tu = Su$, then so must S . Conversely, if S is injective, then there must exist T such that $Tu = Su$. If $\text{null } T_1 = \text{null } T_2$, this implies that the map from the output of T_2 to the output of T_1 is linear, and it exists, so there exists $S \in \mathcal{L}(W)$ such that $T_1 = ST_2$.

32 If $\text{range } T_1 = \text{range } T_2$, then there exists an invertible operator that's linear and maps from V to V such that $T_1 = T_2S$. Since T is a linear map from V to W and $v_1, \dots, v_n$ spans V this also means that $Tv_1, \dots, Tv_n$ spans W . $(S_1S_2 \cdots S_n)(v) = (S_1(v))(S_2(v)) \cdots (S_n(v))$. So if we have another w such that $(S_1S_2 \cdots S_n)(w) = (S_1S_2 \cdots S_n)(v)$, then we must have $S_1(v) = S_1(w)$, etc, so since $S_1, \dots, S_n$ are injective, then $v = w$.

33 We want to show that T is surjective. Obviously, if T is surjective, then U suffices as the input because we really only need one zero, and the rest would be taken care of if we let U be the set consisting of all inputs that don't result in an output of 0, plus one element from the null set, which is $\{0\}$.

Observe that if ST is invertible, then we must be able multiply it by any S^{-1} or T^{-1} , so that we could find S or T . And if S and T are invertible, their determinants are nonzero, so ST 's determinant is also nonzero, so its also invertible.

34 If $ST = I$, then ST is invertible, so $(ST)^{-1} = T^{-1}S^{-1} = I$, so multiplying by T on the left gives $S^{-1} = T$, so multiplying by S on the right gives $TS = I$. If $STU = I$, multiplying by S^{-1} on the left gives $TU = S^{-1}$, then multiplying by U^{-1} on the right gives $T = S^{-1}U^{-1}$, which we know equals $(US)^{-1}$ by problem 1, so $T^{-1} = US$, as desired.

35 Let φ_1 be the functional that maps $\begin{pmatrix} a \\ b \end{pmatrix}$ to a and φ_2 be the functional that maps $\begin{pmatrix} a \\ b \end{pmatrix}$ to b . Since T' represents the linear map from $(\mathbb{R}^2)'$ to $(\mathbb{R}^3)'$, then $T'(\varphi_1)$ and $T'(\varphi_2)$ are linear maps from $\mathbb{R}^2$ to $\mathbb{R}$.

36 ψ_1 would be the functional that maps $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ to a , ψ_2 would be the functional that maps $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ to b , and ψ_3 would be the functional that maps $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ to c . Then, since ψ_1 , ψ_2 , and ψ_3 are parts of a basis, so they are (basis)es though not really, so $T'(\varphi_1)$ and $T'(\varphi_2)$ are linear combinations of ψ_1, ψ_2, ψ_3 .

5.B Eigenvectors and Upper-Triangular Matrices

1

(a) Observe that

$$(I - T)(I + T + \cdots + T^{n-1}) = I - T^n = I,$$

so

$$(I - T)^{-1} = \frac{I}{I - T} = I + T + \cdots + T^{n-1}.$$

(b) We already know that

$$(1 - t)^{-1} = 1 + t + \cdots + t^{n-1}$$

if $t^n = 0$, so we could guess it.

2 We need $T = 2I$, or $T = 3I$, or $T = 4I$, so at least one element of the diagonals must equal 2 or 3 or 4. Thus, by 5.32, $\lambda = 2$ or $\lambda = 3$ or $\lambda = 4$.

3 If $T^2 = I$, we need $T = I$ or $T = -I$. If $T = -I$, then -1 would be an eigenvalue of T , so we need $T = I$.

4 Plugging x^3 into T gives $(Tp)(x^3) = x^6p(x^3) + p''(x^3)$. Then $\varphi(Tp(x^3)) = \varphi(x^6p(x^3) + p''(x^3))$. First, taking the derivative and applying the chain rule gives

$$[x^6p(x^3) + p''(x^3)]' = 6x^5p(x^3) + x^6 \cdot 3x^2p'(x^3) + 3x^2p'''(x^3).$$

Then we plug in $x = 4$ to get an gigantic expression.

5 Assuming that we are considering the transformation of S and S^{-1} on T , we know that

$$p(STS^{-1}) = p(S)p(T)p(S^{-1})$$

and $p(S) = S$ and $p(S^{-1}) = S^{-1}$, so we get $p(STS^{-1}) = Sp(T)S^{-1}$.

6 If $p(z) = a_0 + a_1z + a_2z^2 + \cdots$, then

$$p(T) = a_0 + a_1T + a_2T^2 + \cdots.$$

If U is invariant under T , then it is also invariant under $T^2, T^3, \dots$, so

$$p(T)U = a_0U + a_1(TU) + a_2(T^2U) + \cdots,$$

and since its the sum of invariants, then its also invariant.

7 If 9 is an eigenvalue of T^2 , then $T^2v = 9v$ for some eigenvector v , so $Tw = \pm\sqrt{9}w = \pm 3w$ for some value of v . Thus, 3 or -3 is an eigenvalue of T . For the other direction, if $Tv = 3v$ or $-3v$ for some eigenvector v , then $T^2v^2 = 9v^2$, so v^2 is an eigenvector to 9 for T^2 , as desired.

8 If $T^4 = -1$, then $T^8 = 1$, so a valid transformation is one that rotates a given vector by 45° counterclockwise. The equation for such a vector is

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} = \begin{pmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}.$$

9 Let $p(x) = c_0 + c_1x + c_2x^2 + \cdots + c_nx^n$, so

$$p(T) = c_0 + c_1T + c_2T^2 + \cdots + c_nT^n,$$

so

$$p(T)v = c_0v + c_1Tv + c_2T^2v + \cdots + c_nT^nv.$$

Then if, say, r is a zero of p , then since $p(T)v = 0$, then $Tv = rv$ for some value of v , so r is an eigenvalue of T .

10 Consider the matrix form of T and everything, so this way if you multiply T and v , you get the transformation of v for T (or something like that, how to describe it?). Therefore,

$$p(T)v = c_0v + c_1Tv + c_2T^2v + \cdots + c_nT^nv.$$

If $Tv = \lambda v$, then $T^k v = \lambda^k v$, so we get

$$p(T)v = c_0v + c_1\lambda v + c_2\lambda^2v + \cdots + c_n\lambda^nv = v(c_0 + c_1\lambda + c_2\lambda^2 + \cdots + c_n\lambda^n) = p(\lambda)v.$$

11 Well, since T' maps from $\mathcal{L}(W, \mathbb{R})$ to $\mathcal{L}(V, \mathbb{R})$, then if $T' = 0$, it would also imply that T , which maps from V to W equals 0, and vice versa.

12 Since T is an element of $\mathcal{L}(V, W)$ and T' is an element of $\mathcal{L}(W', V')$, and since these two are distinct and don't have any common elements, it suffices to prove that they have an equal number of elements. Since $(v_1, \dots, v_m)$ encompasses all of V , then the map from the vectors, $\mathbb{F}^m$, to V must be surjective. If $v_1, \dots, v_m$ are linearly independent, this means that the null space has degree 0, and consists of only $\{0\}$.

13 Obviously, if $\varphi(u) = 0$ for some u , then u is automatically an element of null φ . Therefore, if for all $u \in U$, $\varphi(u) = 0 \implies u \in \text{null } \varphi$, then $U \subset \text{null } \varphi$.

14 If $U = \{0\}$, then for the only $u \in U$ is $u = 0$. Obviously, for all $\varphi \in V'$, $\varphi(u) = 0$ for any $u \in U$, namely because there's only one value of u and $\varphi(0) = 0$. Therefore, if $U = \{0\}$, then $U^0 = V'$. Conversely, if for all $\varphi \in V'$, $\varphi(u) = 0$ for every $u \in U$, wouldn't this imply that V' is severely limited? So U can only have one value, namely 0.

15 Observe that if $U^0 = 0$, then only the function $\varphi = 0$ satisfies $\varphi(u) = 0$ for all $u \in U$, so $U = V$. And if $U = V$, then for $\varphi(u) = 0$, we can only have $\varphi = 0$, because then the functionals in V would be constants, so they'd have to equal 0.

16 Consider U^0 , which consists of all $\varphi \in V'$ such that for all $u \in U$, $\varphi(u) = 0$. And we can also consider W^0 , which consists of all $\varphi \in V'$ such that for all $w \in W$, $\varphi(w) = 0$. Note that if $u \in U$, then $u \in W$, so if $\varphi \in W^0$, then we also have $\varphi \in U^0$, so $W^0 \subset U^0$.

17 If $W^0 \subset U^0$, then this implies that for any $\varphi \in V'$, W^0 consists of the φ such that $\varphi(w) = 0$ for all $w \in W$, and U^0 consists of the φ such that $\varphi(u) = 0$ for all $u \in U$. Since $W^0 \subset U^0$, then if we have any φ such that $\varphi(w) = 0$ for all $w \in W$, then $\varphi(u) = 0$ for all $u \in U$. Thus $U \subset W$, as desired.

18 Consider any $x \in U + W$. Automatically, $x \in U$ and $x \in W$. Therefore, if for any $\varphi \in V'$ such that $\varphi \in (U + W)^0$, then for any $x \in U + W \iff x \in U, x \in W$, $\varphi(x) = 0$. Thus, if $\varphi(x) = 0$ for all x , then for the same φ , we also have $\varphi(u) = 0$ for all $u \in U$ and $\varphi(w) = 0$ for all $w \in W$. Hence, $(U + W)^0 \subset U^0 \cap W^0$.

However, $U^0 \cap W^0$ consists of all $\varphi \in V'$ such that $\varphi(u) = 0$ for all $u \in U$ and $\varphi(w) = 0$ for all $w \in W$. If, for any x , $x \in U$ and $x \in W$, then $x \in U + W$, so this also implies that $\varphi(x) = 0$ for all $x \in U + W$, so $U^0 \cap W^0 \subset (U + W)^0$, so $U^0 \cap W^0 = (U + W)^0$, as desired.

19 If $x \in U \cap W$, then $x \in U$ and $x \in W$. Therefore, if $\varphi(x) = 0$ for all $x \in U \cap W$, then $\varphi(x) = 0$ for all $x \in U$, and $\varphi(x) = 0$ for all $x \in W$. Hence, $(U \cap W)^0 \subset U^0 + W^0$.

Additionally, if $\varphi \in U^0 + W^0$, then $\varphi(u) = 0$ for all $u \in U$ and $\varphi(w) = 0$ for all $w \in W$, so $\varphi(x) = 0$ if $x \in U \cap W$. Thus $(U \cap W)^0 = U^0 + W^0$.

20 V is a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$. Prove that T has an invariant subspace of dimension k for each $k = 1, \dots, \dim V$.

5.C Eigenspaces and Diagonal Matrices

1 Consider a basis $u_1, \dots, u_m$ of U , and extend it to a basis $u_1, \dots, u_m, \dots, u_n$ of V , so basically $\dim U = m$ and $\dim V = n$. Now let $\varphi_1, \dots, \varphi_m, \dots, \varphi_n$ be the dual

basis of V' , so then $\varphi_{m+1}, \dots, \varphi_n$ can be a basis of U^0 . Thus $\dim U^0 = n - m$, so $\dim U + \dim U^0 = \dim V$, as desired.

2 If $\varphi(v) = 0$ for every $\varphi \in U^0$ and we are looking for the v , then since $\varphi \in U^0$, we already know that $\varphi(v) = 0$ for all $v \in V$. Thus the set is U .

3

- (a) This is like taking the reverse of the reverse; it goes back to the original. We're looking for the v , then taking the annihilator or something. So this gives $\Gamma = \{v \in V : \varphi(v) = 0 \text{ for every } \varphi \in \Gamma\}^0$.
- (b) If V is finite-dimensional and $\varphi_1, \dots, \varphi_n$ is a basis of V' . Then there must be a basis of V whose dual basis is $\varphi_1, \dots, \varphi_n$ because V' is the dual space of V .
- (c) V' is the set of all linear maps from V to $\mathbb{F}$, and since $\varphi_1, \dots, \varphi_n$ are the functionals such that there exists a basis $v_1, \dots, v_n$ of V such that $\varphi_i(v_j) = 1$ if and only if $i = j$, and 0 otherwise.

4 T' is the linear map from $\mathcal{L}(\mathcal{P}_5(\mathbb{R}), \mathbb{R})$ to $\mathcal{L}(\mathcal{P}_5(\mathbb{R}), \mathbb{R})$. Thus, the range of T consists of all p such that $\varphi(p) = 0$. But $\varphi(p) = p(8)$, so it consists of all p such that $p(8) = 0$.

5 Observe that if both T (which is a linear map from U to V) and S which is a linear map from V to W are invertible, then ST , which is a linear map from U to W is invertible, since it's the product of S and T , and thus $ST = T^{-1}S^{-1}$. This doesn't hold because the sum of two noninvertible operators might just be invertible.

6 If there exists an invertible operator $T \in \mathcal{L}(V)$ such that $Tu = Su$ for every $u \in U$, then this must mean that since T is both injective and surjective, then since $Tu = Su$, then so must S . Conversely, if S is injective, then there must exist T such that $Tu = Su$.

7 We show that everything is equivalent to T is invertible. Obviously, if the columns of $\mathcal{M}(T)$ are linearly independent in $\mathbb{F}^{n,1}$, then it can form a basis (since its linearly independent, and everything that's linearly independent can form a basis), so they are equivalent. Then, if they span $\mathbb{F}^{n,1}$, T is invertible. Similarly, if the rows of $\mathcal{M}(T)$ are linearly independent, it's equivalent to being a basis of $\mathbb{F}^{1,n}$, so it spans $\mathbb{F}^{1,n}$, and vice-versa. Therefore, these 5 statements are equivalent to each other.

8 We already know that Λ maps from V to V'' , so it suffices to show that this transformation is linear. Since $(\Lambda v)(\varphi) = \varphi(v)$, if we have φ_1 and φ_2 , then

$$(\Lambda v)(\varphi_1 + \varphi_2) = (\varphi_1 + \varphi_2)(v) = \varphi_1(v) + \varphi_2(v) = (\Lambda v)(\varphi_1) + (\Lambda v)(\varphi_2).$$

Furthermore,

$$(\Lambda v)(\lambda\varphi) = (\lambda\varphi)(v) = \lambda \cdot \varphi(v) = \lambda(\Lambda v)(\varphi).$$

Therefore, Λ is a linear map from V to V' .

9 We already know that Λ is a linear map from V onto V'' , so we wish to prove that if V is finite-dimensional, $\dim V = \dim V''$. V'' is the set of linear maps from V' to $\mathbb{R}$, so its the set of linear maps from $\mathcal{L}(V, \mathbb{R})$ to $\mathbb{R}$. Thus, we have an isomorphism.

10 $(\mathcal{P}(\mathbb{R}))'$ is the set of linear maps from $\mathcal{P}(\mathbb{R})$ to $\mathbb{R}$. And $\mathbb{R}^\infty$ is the set of all sequences over $\mathbb{R}$. Since a polynomial is basically a sequence of real numbers (take the coefficients), then via a one-to-one correspondence, we're done.

11 U^0 consists of all $\varphi \in V'$ such that $\varphi(u) = 0$ for all $u \in U$. However, we know that $i(u) = u$ for all $u \in U$. Since i' is a map from V' to U' , then it's a map from φ to U' . The null space of i' is the set of i such that $i(u) = 0$ for all $u \in U$, which is equivalent to U^0 .

12 i' is literally a linear map from V' to U' , so of course its range is U' . We already know that i is a linear map from U to V , so i' is a linear map from V' to U' , so i' is

the linear map from $V'/(i') \rightarrow U'$. But, $i' = U^0$ by part (a), so its the linear map from V'/U^0 to U' , as desired.

13 It suffices to show that the null space of π' , which is the set of linear transformations $\in \mathcal{L}(V/U, \mathbb{R})$ that map to 0, has only one element, which is 0. Since V/U is the set of all affine subsets of V parallel to U , then we are done.

14 Since U^0 is defined as the set of $\varphi \in V'$ such that $\varphi(u) = 0$ for all $u \in U$, then since π' maps to functionals in V' , and $\varphi \in V'$ as well, then $\text{range } \pi' = U^0$, as desired.

15 Since $\text{range } \pi' = U^0$, then π' maps from its domain, $(V/U)'$, to U^0 , and thus its a linear map from $(V/U)'$ to U^0 . And, since its injective, and $\dim(V/U)' = \dim U^0$, so its also surjective. Since its both injective and surjective, then its bijective, so its an isomorphism from $(V/U)'$ onto U^0 .

16

- (a) We prove by induction that $T^n(x, y) = (F_{n-1}x + F_ny, F_nx + F_{n+1}y)$. Observe that this is true for $n = 1$. Now assume this is true for some n , so

$$\begin{aligned} T^{n+1}(x, y) &= (F_nx + F_{n+1}y, (F_{n-1}x + F_ny) + (F_nx + F_{n+1}y)) \\ &= (F_nx + F_{n+1}y, F_{n+1}x + F_{n+2}y). \end{aligned}$$

We have now proved our induction, so plugging in $x = 0, y = 1$ gives $T^n(0, 1) = (F_n, F_{n+1})$.

- (b) Let $T(x, y) = (y, x + y) = \lambda(x, y)$, so $y = \lambda x$ and $x + y = \lambda y$. Then $x = (\lambda - 1)y$ and $y = \lambda x = \lambda(\lambda - 1)y$. If $y = 0$, we'd get $x = 0$ as well, so we need $\lambda(\lambda - 1) = 1$. Hence $\lambda^2 - \lambda - 1 = 0$, so $\lambda = \frac{1 \pm \sqrt{5}}{2}$.

- (c) In order to get a basis, we need the vectors to be linearly independent, so they must use different eigenvectors. For $\lambda = \frac{1+\sqrt{5}}{2}$, we let $y = 1$ and plug it in to get $x = (\lambda - 1)y = \frac{\sqrt{5}-1}{2}$. For $\lambda = \frac{1-\sqrt{5}}{2}$, we let $y = -1$ and plug it in to get $x = (\lambda - 1)y = \frac{1+\sqrt{5}}{2}$. Thus we have the basis $\begin{pmatrix} \frac{\sqrt{5}-1}{2} \\ 1 \end{pmatrix}, \begin{pmatrix} \frac{1+\sqrt{5}}{2} \\ -1 \end{pmatrix}$.

- (d) With respect to the basis found in part (c), the matrix of T is

$$\begin{pmatrix} \frac{1+\sqrt{5}}{2} & 0 \\ 0 & \frac{1-\sqrt{5}}{2} \end{pmatrix}.$$

Now observe that

$$T^n = \begin{pmatrix} \left(\frac{1+\sqrt{5}}{2}\right)^n & 0 \\ 0 & \left(\frac{1-\sqrt{5}}{2}\right)^n \end{pmatrix}$$

so when the basis is $(1, 0), (0, 1)$, we get

$$F_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n \right],$$

as desired.

- (e) Well since $\frac{1-\sqrt{5}}{2}$ is less than 1, $\left(\frac{1-\sqrt{5}}{2}\right)^n$ approaches 0, so F_n is the closest to $\frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2}\right)^n$.

Chapter 6: Inner Product Spaces

Linear Algebra Done Right

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6.A Inner Products and Norms

1 Observe that $(\lambda(x_1, x_2), (y_1, y_2)) = |\lambda x_1 y_1| + |\lambda x_2 y_2|$. However, if λ is negative, and x_1, x_2, y_1, y_2 are positive, then this is obviously not equal to $\lambda(|x_1 y_1| + |x_2 y_2|)$.

2 Observe that this function doesn't satisfy definiteness, because if we have the functions $(0, 1, 0)$ and $(0, 1, 0)$, we'll get an output of 0, even though neither input is 0.

3 I believe the problem is incorrect because we have the inner product $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_1 - x_2 y_2$, which is a counterexample. $\langle v, v \rangle = v_1^2 - v_2^2$ may be greater than 0 for some $v \in V$, but its negative for some $v \in V$ as well. Thus its not in the set of inner products with the positivity condition, but its in the set of inner products with replaced condition.

4

(a) We have

$$\begin{aligned}\langle u + v, u - v \rangle &= \langle u, u - v \rangle + \langle v, u - v \rangle \\ &= \langle u, u \rangle - \langle u, v \rangle + \langle v, u \rangle - \langle v, v \rangle \\ &= \langle u, u \rangle - \langle v, v \rangle \\ &= \|u\|^2 - \|v\|^2.\end{aligned}$$

(b) If $\|u\| = \|v\|$, then $\langle u + v, u - v \rangle = \|u\|^2 - \|v\|^2 = 0$, so $u + v$ and $u - v$ are orthogonal.

(c) If u and v are two adjacent sides of a rhombus, then the diagonals are $u + v$ and $u - v$ in some order. It's a rhombus, so $\|u\| = \|v\|$, so $u + v$ and $u - v$ are orthogonal by part (b), so the diagonals are perpendicular as desired.

5 By 5.6, $T - \lambda I$ being not invertible is equivalent to λ being an eigenvalue, so λ not being an eigenvalue is equivalent to $T - \lambda I$ being invertible. $\sqrt{2}$ is obviously not an eigenvalue because if $\|Tv\| \leq \|v\|$, then $\|Tv\| < \sqrt{2}\|v\|$, so $Tv < \sqrt{2}v$, so we're done.

6 Observe that if $\langle u, v \rangle = 0$, then u and v are orthogonal, so no matter what value a is, av will be orthogonal to u . $u + av$ is the hypotenuse of a right triangle with legs u and av , so $\|u + av\| \geq \|u\|$ by the Pythagorean Theorem.

Now, we prove the other way around. Suppose for the sake of contradiction that $\|u\| \leq \|u + av\|$ for all $a \in \mathbb{F}$, and u and v are not orthogonal. Now we will find a value of $a \in \mathbb{F}$ such that $\|u\| > \|u + av\|$. First, draw the vectors u and v , starting at the origin. Make sure their starting points are the origin. Then draw a circle at the origin with radius $\|u\|$, so the head of u is on the circle. Then draw a line through the head of u parallel to v , so that it intersects the circle at some point. Note that if u and v were

orthogonal, then it would be tangent. Anyway, if it intersects the circle at points, then if you connect the origin to a point on this line, it will be $u + av$ for some value of a , but its norm would be less than $\|u\|$, as desired.

7 First, we prove that if $\|u\| = \|v\|$, then $\|au + bv\| = \|bu + av\|$. Observe that $au + bv$ is the diagonal of a parallelogram with sides au and bv , and $bu + av$ is the diagonal of a parallelogram with sides bu and av . Since $\|u\| = \|v\|$, the parallelogram with sides bu and av is just the reflection of the parallelogram with sides au and bv over the angle bisector of u and v . Hence $\|au + bv\| = \|bu + av\|$.

Also, if $\|au + bv\| = \|bu + av\|$ for all $a, b \in \mathbb{R}$, then letting $a = 1$ and $b = 0$ gives $\|u\| = \|v\|$.

8 Since $\sqrt{\langle u, v \rangle} = \|v\| = \sqrt{\langle v, v \rangle}$, then $u = v$.

9 This is made apparent by Cauchy-Schwarz, since $|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$.

10 Let $u = (a, 3a)$ for some $a \in \mathbb{R}$. Since v is orthogonal to $(1, 3)$, then $v = (-3b, b)$ for some b . Then $u + v = (a - 3b, 3a + b) = (1, 2)$, so $a = \frac{7}{10}$ and $b = -\frac{1}{10}$. Then $u = (\frac{7}{10}, \frac{21}{10})$ and $v = (\frac{3}{10}, -\frac{1}{10})$.

11 We have Cauchy-Schwarz! If we let $u = \langle x_1, \dots, x_n \rangle$ and $v = \langle y_1, \dots, y_n \rangle$, then we know that

$$\langle u, v \rangle = x_1 y_1 + \dots + x_n y_n$$

and

$$\|u\| \cdot \|v\| = \sqrt{x_1^2 + \dots + x_n^2} \cdot \sqrt{y_1^2 + \dots + y_n^2}.$$

Thus we get

$$|x_1 y_1 + \dots + x_n y_n| \leq \sqrt{x_1^2 + \dots + x_n^2} \cdot \sqrt{y_1^2 + \dots + y_n^2},$$

and squaring both sides yields the more common form of Cauchy-Schwarz, which is

$$(x_1 y_1 + \dots + x_n y_n)^2 \leq (x_1^2 + \dots + x_n^2)(y_1^2 + \dots + y_n^2).$$

Now putting this to use with $x_1 = a$, etc, and $y_1 = \frac{1}{a}$, etc, gives

$$\sum_{\text{cyc}} a \cdot \sum_{\text{cyc}} \frac{1}{a} \geq \left(\sum_{\text{cyc}} a \cdot \frac{1}{a} \right)^2 = 4^2 = 16.$$

12 We have Cauchy-Schwarz!

$$\sum_{\text{cyc}} 1^2 \cdot \sum_{\text{cyc}} x_1^2 \geq \left(\sum_{\text{cyc}} \sqrt{1^2 \cdot x_1^2} \right)^2,$$

so

$$n(x_1^2 + \dots + x_n^2) \geq (x_1 + \dots + x_n)^2,$$

as desired.

13 $\langle \cdot, \cdot \rangle$ must have been the dot product all along! Anyway, consider the triangle with sides u , v , and $u - v$. We apply the law of cosines on $u - v$ to get

$$\|u - v\|^2 = \|u\|^2 + \|v\|^2 - 2\|u\| \cdot \|v\| \cos \theta.$$

Since $\|u - v\|^2 = \|u\|^2 + \|v\|^2 - 2\langle u, v \rangle$, then $\langle u, v \rangle = \|u\| \cdot \|v\| \cos \theta$, as desired.

14 The arccos function is only defined when its argument is between -1 and 1 inclusive. Since the Cauchy-Schwarz inequality shows that $|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$, or equivalently, $-1 \leq \frac{\langle u, v \rangle}{\|u\| \cdot \|v\|} \leq 1$, we're done.

15 Using Cauchy-Schwarz gives

$$\left(\sum_{j=1}^n ja_j^2\right) \left(\sum_{j=1}^n \frac{b_j^2}{j}\right) \geq \left(\sum_{j=1}^n \sqrt{ja_j^2 \cdot \frac{b_j^2}{j}}\right)^2 = \left(\sum_{j=1}^n a_j b_j\right)^2,$$

as desired.

16

$$\begin{aligned} \|u + v\|^2 &= \|u\|^2 + \|v\|^2 + 2\langle u, v \rangle = 16 \implies \|v\|^2 + 2\langle u, v \rangle = 7 \\ \|u - v\|^2 &= \|u\|^2 + \|v\|^2 - 2\langle u, v \rangle = 36 \implies \|v\|^2 - 2\langle u, v \rangle = 27. \end{aligned}$$

Adding gives $2\|v\|^2 = 34 \implies \|v\|^2 = 17 \implies \|v\| = \sqrt{17}$.

17 If, say, u equals $(-a, b)$, where both a and b are positive, then $\|u\| = b$. Then $\lambda u = (-\lambda a, \lambda b)$, and if λ is negative, then $\|\lambda u\| = -\lambda a$, which is not equal to $|\lambda| \cdot \|u\| = -\lambda b$.

18 Note that if p is odd, then letting $y = -x$ gives $\|(x, y)\| = 0$ even though $(x, y) \neq 0$, so p can't be odd. We know that $p = 2$ works (its the standard dot product), and we also know that $\|(x, y)\| = \sqrt{\langle (x, y), (x, y) \rangle}$, so we must have $p = 2$.

19 We know that

$$\begin{aligned} \|u + v\|^2 &= \|u\|^2 + \|v\|^2 + 2\langle u, v \rangle \\ \|u - v\|^2 &= \|u\|^2 + \|v\|^2 - 2\langle u, v \rangle. \end{aligned}$$

Thus subtracting the two equations gives

$$4\langle u, v \rangle = \|u + v\|^2 - \|u - v\|^2 \implies \langle u, v \rangle = \frac{\|u + v\|^2 - \|u - v\|^2}{4},$$

as desired.

20 We know that $\|u + v\|^2 = \|u\|^2 + \|v\|^2 + \langle u, v \rangle + \langle v, u \rangle$ and $\|u - v\|^2 = \|u\|^2 + \|v\|^2 - \langle u, v \rangle - \langle v, u \rangle$, so $\|u + v\|^2 - \|u - v\|^2 = 2\langle u, v \rangle + 2\langle v, u \rangle$. Now

$$\|u + iv\|^2 = \langle u + iv, u + iv \rangle = \langle u, u \rangle + \langle iv, iv \rangle + \langle u, iv \rangle + \langle iv, u \rangle.$$

Since

$$\langle iv, iv \rangle = \bar{i}\langle iv, v \rangle = -i\langle iv, v \rangle = -i \cdot i\langle v, v \rangle = \langle v, v \rangle = \|v\|^2,$$

then

$$\|u + iv\|^2 = \|u\|^2 + \|v\|^2 - i\langle u, v \rangle + i\langle v, u \rangle.$$

Similarly,

$$\begin{aligned} \|u - iv\|^2 &= \langle u - iv, u - iv \rangle \\ &= \langle u, u \rangle + \langle iv, iv \rangle + \langle -iv, u \rangle + \langle u, -iv \rangle \\ &= \|u\|^2 + \|v\|^2 - i\langle v, u \rangle + i\langle u, v \rangle. \end{aligned}$$

Thus

$$\begin{aligned} \|u + iv\|^2 - \|u - iv\|^2 &= 2i\langle v, u \rangle - 2i\langle u, v \rangle \\ \implies \|u + iv\|^2 i - \|u - iv\|^2 i &= 2\langle u, v \rangle - 2\langle v, u \rangle. \end{aligned}$$

Therefore

$$\frac{\|u + v\|^2 - \|u - v\|^2 + \|u + iv\|^2 i - \|u - iv\|^2 i}{4} = \langle u, v \rangle,$$

as desired.

21 Note that the norm $\| \cdot \|$ can obviously be one from an inner product, just let it. Because this norm satisfies all the properties that an inner product norm has, then it must be an inner product norm.

22 We want to show that if $a_1, \dots, a_n \in \mathbb{R}$, then

$$\left(\frac{a_1 + \dots + a_n}{n} \right)^2 \leq \frac{a_1^2 + \dots + a_n^2}{n}.$$

Multiplying both sides by n^2 gives

$$(a_1 + \dots + a_n)^2 \leq n(a_1^2 + \dots + a_n^2),$$

which we proved in problem 12, and vice versa, so we're done.

23. We know that $\mathcal{M}(T)$ is defined as the set of numbers $a_{i,j}$ for $1 \leq i \leq m$, $1 \leq j \leq n$ such that for all j ,

$$w_j = a_{1,j}v_1 + a_{2,j}v_2 + \dots + a_{m,j}v_m.$$

So, if we add w_{s_j} to w_{t_j} (these are the respective bases of S and T), and let a and b denote the values of the entries in the matrices, we get

$$\mathcal{M}(S + T) = (a_{1,i} + b_{1,i})v_1 + \dots + (a_{m,i} + b_{m,i})v_m.$$

24 $\langle v, v \rangle_1 = \langle Sv, Sv \rangle$, which we can let equal $\|v\|^2$. Since S is injective and its linear, we also have the properties of definiteness, additivity in first slot, and homogeneity in first slot. Since $\langle Su, Sv \rangle$ satisfies conjugate symmetry, so does $\langle \cdot, \cdot \rangle_1$, so we're done.

25 Because S is not injective, then the null space of S isn't limited to $\{0\}$. Therefore, $\langle 0, 0 \rangle_1$ isn't the only inner product that satisfies $\langle \cdot, \cdot \rangle_1 = 0$. So it doesn't satisfy definiteness.

26

- (a) We know that the inner product is some combination of the products of corresponding elements in the vector. Since $f(t)$, $g(t)$, $f'(t)$, and $g'(t)$ are all vectors, then the expression is defined. Then if we let

$$f(t) = \begin{pmatrix} a_1(t) \\ \vdots \\ a_n(t) \end{pmatrix} \quad \text{and} \quad g(t) = \begin{pmatrix} b_1(t) \\ \vdots \\ b_n(t) \end{pmatrix}$$

then

$$f'(t) = \begin{pmatrix} a'_1(t) \\ \vdots \\ a'_n(t) \end{pmatrix} \quad \text{and} \quad g'(t) = \begin{pmatrix} b'_1(t) \\ \vdots \\ b'_n(t) \end{pmatrix}.$$

We know that by the product rule, $(a_i(t)b_i(t))' = a'_i(t)b_i(t) + a_i(t)b'_i(t)$, and since the formula for $\langle \cdot, \cdot \rangle$ is the same, then

$$\langle f(t), g(t) \rangle' = \langle f'(t), g(t) \rangle + \langle f(t), g'(t) \rangle,$$

as desired.

- (b) By part (a),

$$\langle f(t), f(t) \rangle' = \langle f'(t), f(t) \rangle + \langle f(t), f'(t) \rangle = 2\langle f'(t), f(t) \rangle.$$

Since $\langle f(t), f(t) \rangle = \|f(t)\|^2 = c^2$, which is constant, then $\langle f(t), f(t) \rangle' = 0$, so $\langle f'(t), f(t) \rangle = 0$.

- (c) Since $f(t)$ is constant, then the head of $f(t)$ must lie on a circle of radius c centered at the origin. $f'(t)$ is the tangent line to the circle at that point. Since $\langle f'(t), f(t) \rangle = 0$, then $f'(t)$ is orthogonal to $f(t)$, so this implies that the tangent line to a circle at a point is perpendicular to the radius vector.

27 Observe that $w - \frac{1}{2}(u + v) = \frac{(w-u) + (w-v)}{2}$ and $u - v = (w - v) - (w - u)$, so

$$\|w - \frac{1}{2}(u + v)\|^2 = \left\| \frac{(w-u) + (w-v)}{2} \right\|^2 = \frac{\|w-u\|^2 + \|w-v\|^2 + 2\langle w-u, w-v \rangle}{4}.$$

We also know that

$$\frac{\|u-v\|^2}{4} = \frac{\|(w-v) - (w-u)\|^2}{4} = \frac{\|w-v\|^2 + \|w-u\|^2 - 2\langle w-v, w-u \rangle}{4}.$$

So

$$\|w - \frac{1}{2}(u + v)\|^2 + \frac{\|u-v\|^2}{4} = \frac{\|w-u\|^2 + \|w-v\|^2}{4},$$

so rearranging gives the desired equality.

28 For the sake of contradiction suppose there exists another $v \in \mathbb{C}$ such that $\|w-v\|$ is minimized and $u \neq v$. We get $\|w-u\| \leq \|w-v\| \leq \|w-u\|$, so $\|w-u\| = \|w-v\|$. Then by the previous problem, $\frac{1}{2}(u+v)$ exists, and

$$\|w - \frac{1}{2}(u + v)\|^2 = \|w-u\|^2 - \frac{\|u-v\|^2}{4} < \|w-u\|^2.$$

This is a contradiction because we assumed $\|w-u\|$ was the smallest, so we are done.

29

- (a) Since this is the norm we're dealing with, obviously $d(u, v) \geq 0$. Also, if $\|u-v\| = 0$, then $u-v=0$, so $u=v$. $\|u-v\| = \|v-u\|$, and since $\|u-v\|$ is equal to the length of the line segment connecting u and v , it obviously also satisfies triangle inequality. $\|u-v\| + \|v-w\| \geq \|u-w\|$.

- (b) If we have

$$Tv_i = a_{1,i}v_1 + \cdots + a_{m,i}v_i,$$

then if we multiply T times λ , we get

$$\lambda a_{1,i}v_1 + \cdots + \lambda a_{m,i}v_i = a_{1,i}(\lambda v_1) + \cdots + a_{m,i}(\lambda v_i)$$

as desired.

- (c) Observe that we wish to prove that when we multiply a matrix by a vector, we get a combination of all of the previous vectors and put them together. This is clear because we already have a collection of vectors to start with. A single value times matrix put together gives a vector of values times matrix.

30 A reasonable guess is that the desired harmonic polynomial p is of the form $q + (1 - \|x\|^2)r$ for some polynomial r . It suffices to prove that r exists. Define an operator T on the vector space by $Tr = \Delta((1 - \|x\|^2)r)$. Observe that T is injective, so T is surjective, and we're done.

31 Consider a triangle with u and v as sides, and suppose they are the upper two in the given diagram. Then $\|u\| = a$ and $\|v\| = b$, $\|\frac{u+v}{2}\| = d$, and $\|u-v\| = c$. It suffices to show that

$$\|u\|^2 + \|v\|^2 = \frac{1}{2}\|u-v\|^2 + 2\left\|\frac{u+v}{2}\right\|^2 = \frac{\|u-v\|^2 + \|u+v\|^2}{2}.$$

Since $\|u-v\|^2 = \|u\|^2 + \|v\|^2 - 2\langle u, v \rangle$ and $\|u+v\|^2 = \|u\|^2 + \|v\|^2 + 2\langle u, v \rangle$, then we're done.

6.B Orthonormal Bases

1

- (a) First, note that they are orthonormal because $(\cos \theta, \sin \theta) \cdot (-\sin \theta, \cos \theta) = -\cos \theta \sin \theta + \sin \theta \cos \theta = 0$ and $(\cos \theta, \sin \theta) \cdot (\sin \theta, -\cos \theta) = \cos \theta \sin \theta - \sin \theta \cos \theta = 0$. Then note that because they're perpendicular, then they're not scalar multiples of the other, so they are also bases of $\mathbb{R}^2$.
- (b) Since they're orthonormal bases, then they must consist of two vectors that lie on the unit circle that are perpendicular to each other. So if we let one vector be $(\cos \theta, \sin \theta)$, then the other must be either $(-\sin \theta, \cos \theta)$ or $(\sin \theta, -\cos \theta)$.

2 Consider $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and $C = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$. Then

$$AC = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$$

and

$$CA = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$$

as desired.

3 Observe that if $(AB)C$ makes sense, then if A is $m \times n$, then B must be $n \times p$ for some p , so AB is $m \times p$, so C is $p \times q$ for some q . Thus BC is $n \times q$, so $A(BC)$ is $m \times q$, which is defined.

4 We already know that the entry in any given row and column of A^2 is equal to the sum of the products of $a_{i,j}$, so continuing with A^3 gives $a_{j,p}a_{p,r}a_{r,k}$.

5 Observe that if both T (which is a linear map from U to V) and S which is a linear map from V to W are invertible, then ST , which is a linear map from U to W is invertible, since it's the product of S and T , and thus $ST = T^{-1}S^{-1}$. This doesn't hold because the sum of two noninvertible operators might just be invertible.

6 If there exists an invertible operator $T \in \mathcal{L}(V)$ such that $Tu = Su$ for every $u \in U$, then this must mean that since T is both injective and surjective, then since $Tu = Su$, then so must S . Conversely, if S is injective, then there must exist T such that $Tu = Su$.

7 If $\text{null } T_1 = \text{null } T_2$, this implies that the map from the output of T_2 to the output of T_1 is linear, and it exists, so there exists $S \in \mathcal{L}(W)$ such that $T_1 = ST_2$.

8 If $\text{range } T_1 = \text{range } T_2$, then there exists an invertible operator that's linear and maps from V to V such that $T_1 = T_2S$.

9 Then one of the vectors in the Gram-Schmidt representation will be equal to 0.

9 Observe that if ST is invertible, then we must be able multiply it by any S^{-1} or T^{-1} , so that we could find S or T . And if S and T are invertible, their determinants are nonzero, so ST 's determinant is also nonzero, so it's also invertible.

10 If $ST = I$, then ST is invertible, so $(ST)^{-1} = T^{-1}S^{-1} = I$, so multiplying by T on the left gives $S^{-1} = T$, so multiplying by S on the right gives $TS = I$.

11 If $STU = I$, multiplying by S^{-1} on the left gives $TU = S^{-1}$, then multiplying by U^{-1} on the right gives $T = S^{-1}U^{-1}$, which we know equals $(US)^{-1}$ by problem 1, so $T^{-1} = US$, as desired.

12 If V isn't finite-dimensional, then S , T , and U might not necessarily have inverses, so there goes the proof.

13 If R , S , and T are all linear maps from V to V itself, since RST is plugging the output of T into S and plugging this into R , then since T is surjective, then S is injective.

14 Suppose $e_1, \dots, e_n$ is an orthonormal basis of V and $v_1, \dots, v_n$ are vectors in V such that

$$\|e_j - v_j\| < \frac{1}{\sqrt{n}}$$

for each j . Then $v_1, \dots, v_n$ is a basis of V . Observe that if the only solution if $c_1 = \dots = c_n = 0$ is all 0's, then this means that solutions exist, and its linearly independent ($A_{\cdot k}$ for all k) so it can also act as a basis, so there will have to exist x_k . And if there exist x_k as solutions to the c , then it's a basis, so it's linearly independent, so $x_1 = \dots = x_n = 0$ is the only solution if all the c 's equal 0.

15 Essentially, we want to prove that there doesn't exist g such that for every f that's continuous on $[-1, 1]$,

$$\varphi(f) = \langle f, g \rangle = \int_{-1}^1 f(x)g(x) dx.$$

Observe that this doesn't occur because $\varphi(f) = \langle f, g \rangle$ might equal 0 at some point, so $f(0) < 0$.

16 Consider the map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^2$. Then if we map $T(a, b, c, d, e)$ to $(a, b, 0, 0, 0)$, then the range has dimension 2, and the null space consists of all quintuplets of the form $(0, 0, c, d, e)$, which has dimension 3. Suppose $S : X \rightarrow Y$ and $T : U \rightarrow V$. Then if we have $x \in X$, which maps to a $y \in Y$, then $y \in \text{null } T$, so $T(y) = T(S(x)) = 0$, so $TS = 0$, so $(ST)^2 = 0$.

17

- (a) Since $(v_1, \dots, v_m)$ encompasses all of V , then the map from the vectors, $\mathbb{F}^m$, to V must be surjective. If $v_1, \dots, v_m$ are linearly independent, this means that the null space has degree 0, and consists of only $\{0\}$.
- (b) Since we know that $\dim \text{null } T + \dim \text{range } T = \dim V$ but $\dim V = 5$ and $\dim \text{null } T > 2$ and $\dim \text{range } T = 4$, which is impossible. How about something like $T : (a, b, c, d) \rightarrow (a, b, 0, 0)$. Then the range has dimension 2 but the null space also has dimension 2.
- (c) Because we know that $\dim V = \dim \text{range } T + \dim \text{null } T$, but $\dim V = 5$, which is odd, so we can't split it evenly. Observe that since $\dim \text{range } T \geq \dim V$, but $\dim \text{null } T \geq 1$ because T is not injective, then it is impossible for $\dim \text{range } T + \dim \text{null } T$ to equal $\dim V$, as desired.
- (d) Since $\mathbb{F} = \mathbb{R}$ and V is finite-dimensional, then φ is an isomorphism from V to V' because $\dim V = \dim V'$. Since $v_1, \dots, v_n$ is linearly independent, then since T is injective, we also know that the null space of T equals 0, so $Tv_1, \dots, Tv_n$ is also linearly independent because we know that T is a linear map, so $Tv + Tw = T(v + w)$. And since its injective, each v maps to a distinct output, so we can look at the inputs instead, which are linearly independent.

6.C Orthogonal Complements and Minimization Problems

1 $(S_1 S_2 \dots S_n)(v) = (S_1(v))(S_2(v)) \dots (S_n(v))$. So if we have another w such that $(S_1 S_2 \dots S_n)(w) = (S_1 S_2 \dots S_n)(v)$, then we must have $S_1(v) = S_1(w)$, etc, so since $S_1, \dots, S_n$ are injective, then $v = w$.

2 We want to show that T is surjective. Obviously, if T is surjective, then U suffices as the input because we really only need one zero, and the rest would be taken care of if

we let U be the set consisting of all inputs that don't result in an output of 0, plus one element from the null set, which is $\{0\}$.

3 Observe that the null space consists of points of the form $(5a, a, 7b, b)$, which has dimension 2. Since $\dim \text{null } T + \dim \text{range } T = \dim V = 4$, then $\dim \text{range } T = 2 = \dim W$, so T is surjective, as desired.

4 Since $\text{null } T = U$, then since U is 3-dimensional, then $\dim \text{null } T = 3$ as well. $\dim \text{null } T + \dim \text{range } T = \dim V = 8$, so $\dim \text{range } T = 5 = \dim W$, so T is surjective, as desired.

5 So the null space consists of quintuplets of the form $(3a, a, b, b, b)$, which has dimension 2. Since $\dim \text{range } T = \dim V - \dim \text{null } T = 5 - 2 = 3$, then $\dim \text{range } T > \dim W$, which is impossible.

6 Suppose V is finite-dimensional and $P \in \mathcal{L}(V)$

7 Consider the map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^2$. Then if we map $T(a, b, c, d, e)$ to $(a, b, 0, 0, 0)$, then the range has dimension 2, and the null space consists of all quintuplets of the form $(0, 0, c, d, e)$, which has dimension 3.

8 Suppose $S : X \rightarrow Y$ and $T : U \rightarrow V$. Then if we have $x \in X$, which maps to a $y \in Y$, then $y \in \text{null } T$, so $T(y) = T(S(x)) = 0$, so $TS = 0$, so $(ST)^2 = 0$.

9 Since $(v_1, \dots, v_m)$ encompasses all of V , then the map from the vectors, $\mathbb{F}^m$, to V must be surjective. If $v_1, \dots, v_m$ are linearly independent, this means that the null space has degree 0, and consists of only $\{0\}$.

10 Since we know that $\dim \text{null } T + \dim \text{range } T = \dim V$ but $\dim V = 5$ and $\dim \text{null } T > 2$ and $\dim \text{range } T = 4$, which is impossible.

11 Observe that if both T (which is a linear map from U to V) and S which is a linear map from V to W are invertible, then ST , which is a linear map from U to W is invertible, since it's the product of S and T , and thus $ST = T^{-1}S^{-1}$.

12 This doesn't hold because the sum of two noninvertible operators might just be invertible. If there exists an invertible operator $T \in \mathcal{L}(V)$ such that $Tu = Su$ for every $u \in U$, then this must mean that since T is both injective and surjective, then since $Tu = Su$, then so must S . Conversely, if S is injective, then there must exist T such that $Tu = Su$.

13 If $\text{null } T_1 = \text{null } T_2$, this implies that the map from the output of T_2 to the output of T_1 is linear, and it exists, so there exists $S \in \mathcal{L}(W)$ such that $T_1 = ST_2$. If $\text{range } T_1 = \text{range } T_2$, then there exists an invertible operator that's linear and maps from V to V such that $T_1 = T_2S$. Since T is a linear map from V to W and $v_1, \dots, v_n$ spans V this also means that $Tv_1, \dots, Tv_n$ spans W .

14

(a) $(S_1 S_2 \cdots S_n)(v) = (S_1(v))(S_2(v)) \cdots (S_n(v))$. So if we have another w such that $(S_1 S_2 \cdots S_n)(w) = (S_1 S_2 \cdots S_n)(v)$, then we must have $S_1(v) = S_1(w)$, etc, so since $S_1, \dots, S_n$ are injective, then $v = w$.

(b) We want to show that T is surjective. Obviously, if T is surjective, then U suffices as the input because we really only need one zero, and the rest would be taken care of if we let U be the set consisting of all inputs that don't result in an output of 0, plus one element from the null set, which is $\{0\}$.

Chapter 7: Operators on Inner Product Spaces

Linear Algebra Done Right

Carissa Zeng

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7.A Self-Adjoint and Normal Operators

1 Since $v_1, \dots, v_n$ is linearly independent, then since T is injective, we also know that the null space of T equals 0, so $Tv_1, \dots, Tv_n$ is also linearly independent because we know that T is a linear map, so $Tv + Tw = T(v + w)$. And since its injective, each v maps to a distinct output, so we can look at the inputs instead, which are linearly independent.

2 Since T is a linear map from V to W and $v_1, \dots, v_n$ spans V this also means that $Tv_1, \dots, Tv_n$ spans W . $(S_1 S_2 \cdots S_n)(v) = (S_1(v))(S_2(v)) \cdots (S_n(v))$. So if we have another w such that $(S_1 S_2 \cdots S_n)(w) = (S_1 S_2 \cdots S_n)(v)$, then we must have $S_1(v) = S_1(w)$, etc, so since $S_1, \dots, S_n$ are injective, then $v = w$.

3 We want to show that T is surjective. Obviously, if T is surjective, then U suffices as the input because we really only need one zero, and the rest would be taken care of if we let U be the set consisting of all inputs that don't result in an output of 0, plus one element from the null set, which is $\{0\}$.

4

(a) Observe that the null space consists of points of the form $(5a, a, 7b, b)$, which has dimension 2. Since $\dim \text{null } T + \dim \text{range } T = \dim V = 4$, then $\dim \text{range } T = 2 \dim W$, so T is surjective, as desired.

(b) Since $\text{null } T = U$, then since U is 3-dimensional, then $\dim \text{null } T = 3$ as well. $\dim \text{null } T + \dim \text{range } T = \dim V = 8$, so $\dim \text{range } T = 5 = \dim W$, so T is surjective. So the null space consists of quintuplets of the form $(3a, a, b, b, b)$, which has dimension 2. Since $\dim \text{range } T = \dim V - \dim \text{null } T = 5 - 2 = 3$, then $\dim \text{range } T > \dim W$, which is impossible.

5 Thus there exists values such that

$$Tv = \varphi_1(v)w_1 + \cdots + \varphi_m(v)w_m.$$

Consider the map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^2$. Then if we map $T(a, b, c, d, e)$ to $(a, b, 0, 0, 0)$, then the range has dimension 2, and the null space consists of all quintuplets of the form $(0, 0, c, d, e)$, which has dimension 3. Suppose $S : X \rightarrow Y$ and $T : U \rightarrow V$. Then if we have $x \in X$, which maps to a $y \in Y$, then $y \in \text{null } T$, so $T(y) = T(S(x)) = 0$, so $TS = 0$, so $(ST)^2 = 0$.

6

- (a) Since $(v_1, \dots, v_m)$ encompasses all of V , then the map from the vectors, $\mathbb{F}^m$, to V must be surjective. If $v_1, \dots, v_m$ are linearly independent, this means that the null space has degree 0, and consists of only $\{0\}$. Since we know that $\dim \text{null } T + \dim \text{range } T = \dim V$ but $\dim V = 5$ and $\dim \text{null } T > 2$ and $\dim \text{range } T = 4$, which is impossible.
- (b) How about something like $T : (a, b, c, d) \rightarrow (a, b, 0, 0)$. Then the range has dimension 2 but the null space also has dimension 2. Because we know that $\dim V = \dim \text{range } T + \dim \text{null } T$, but $\dim V = 5$, which is odd, so we can't split it evenly. Observe that since $\dim \text{range } T \geq \dim V$, but $\dim \text{null } T \geq 1$ because T is not injective, then it is impossible for $\dim \text{range } T + \dim \text{null } T$ to equal $\dim V$, as desired.

7 Because T is not surjective, this means that $\dim \text{range } T < \dim W \leq \dim V$ and $\dim \text{null } T$. Then $\dim \text{range } T < \dim W \leq \dim V$. Hence, $\dim \text{null } T + \dim \text{range } T > \dim V$, so we're done.

Since $v_1, \dots, v_n$ is linearly independent, then since T is injective, we also know that the null space of T equals 0, so $Tv_1, \dots, Tv_n$ is also linearly independent because we know that T is a linear map, so $Tv + Tw = T(v + w)$. And since its injective, each v maps to a distinct output, so we can look at the inputs instead, which are linearly independent.

8 Since T is a linear map from V to W and $v_1, \dots, v_n$ spans V this also means that $Tv_1, \dots, Tv_n$ spans W . $(S_1 S_2 \cdots S_n)(v) = (S_1(v))(S_2(v)) \cdots (S_n(v))$. So if we have another w such that $(S_1 S_2 \cdots S_n)(w) = (S_1 S_2 \cdots S_n)(v)$, then we must have $S_1(v) = S_1(w)$, etc, so since $S_1, \dots, S_n$ are injective, then $v = w$.

9 We want to show that T is surjective. Obviously, if T is surjective, then U suffices as the input because we really only need one zero, and the rest would be taken care of if we let U be the set consisting of all inputs that don't result in an output of 0, plus one element from the null set, which is $\{0\}$.

10 Observe that the null space consists of points of the form $(5a, a, 7b, b)$, which has dimension 2. Since $\dim \text{null } T + \dim \text{range } T = \dim V = 4$, then $\dim \text{range } T = 2$ $\dim W$, so T is surjective, as desired. Since $\text{null } T = U$, then since U is 3-dimensional, then $\dim \text{null } T = 3$ as well. $\dim \text{null } T + \dim \text{range } T = \dim V = 8$, so $\dim \text{range } T = 5 = \dim W$, so T is surjective, as desired.

11 So the null space consists of quintuplets of the form $(3a, a, b, b, b)$, which has dimension 2. Since $\dim \text{range } T = \dim V - \dim \text{null } T = 5 - 2 = 3$, then $\dim \text{range } T > \dim W$, which is impossible. Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. There exists a basis $w_1, \dots, w_n$ of W such that all the entries in the first column of $\mathcal{M}(T)$ (with respect to the bases $v_1, \dots, v_m$ and $w_1, \dots, w_n$) are 0 except for possibly T as a linear map from V to W such that there is nothing except possibly, a 0 in the first row and first column.

12 If we have the matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

In this case, we have a matrix with a 1's along the diagonal, meaning places where the number of rows and number of columns are equal, and 0 everywhere else.

13 If $\dim \text{range } T = 1$, this means that $\dim \text{null } T = \dim V - 1$. Therefore there needs to exist a basis of V and a basis of W such that all entries of $\mathcal{M}(T)$ equal 1, because this would We know that $\mathcal{M}(T)$ is defined as the set of numbers $a_{i,j}$ for $1 \leq i \leq m$, $1 \leq j \leq n$ such that for all j ,

$$w_j = a_{1,j}v_1 + a_{2,j}v_2 + \cdots + a_{m,j}v_m.$$

14 So, if we add w_{s_j} to w_{t_j} (these are the respective bases of S and T), and let a and b denote the values of the entries in the matrices, we get

$$\mathcal{M}(S + T) = (a_{1,i} + b_{1,i})v_1 + \cdots + (a_{m,i} + b_{m,i})v_m$$

and

$$\begin{aligned}\mathcal{M}(S) + \mathcal{M}(T) &= [a_{1,i}v_1 + \cdots + a_{m,i}v_m] \\ &\quad + [b_{1,i}v_1 + \cdots + b_{m,i}v_m] \\ &= (a_{1,i} + b_{1,i})v_1 + \cdots + (a_{m,i} + b_{m,i})v_m\end{aligned}$$

as desired.

15

(a) If we have

$$Tv_i = a_{1,i}v_1 + \cdots + a_{m,i}v_m,$$

then if we multiply T times λ , we get

$$\lambda a_{1,i}v_1 + \cdots + \lambda a_{m,i}v_m = a_{1,i}(\lambda v_1) + \cdots + a_{m,i}(\lambda v_m)$$

as desired.

(b) Observe that we wish to prove that when we multiply a matrix by a vector, we get a combination of all of the previous vectors and put them together. This is clear because we already have a collection of vectors to start with. A single value times matrix put together gives a vector of values times matrix.

16 Observe that if both T (which is a linear map from U to V) and S which is a linear map from V to W are invertible, then ST , which is a linear map from U to W is invertible, since it's the product of S and T , and thus $ST = T^{-1}S^{-1}$. This doesn't hold because the sum of two noninvertible operators might just be invertible.

17 Observe that if $(AB)C$ makes sense, then if A is $m \times n$, then B must be $n \times p$ for some p , so AB is $m \times p$, so C is $p \times q$ for some q . Thus BC is $n \times q$, so $A(BC)$ is $m \times q$, which is defined.

18 We already know that the entry in any given row and column of A^2 is equal to the sum of the products of $a_{i,j}$, so continuing with A^3 gives $a_{j,p}a_{p,r}a_{r,k}$.

If there exists an invertible operator $T \in \mathcal{L}(V)$ such that $Tu = Su$ for every $u \in U$, then this must mean that since T is both injective and surjective, then since $Tu = Su$, then so must S . Conversely, if S is injective, then there must exist T such that $Tu = Su$.

19 If $\text{null } T_1 = \text{null } T_2$, this implies that the map from the output of T_2 to the output of T_1 is linear, and it exists, so there exists $S \in \mathcal{L}(W)$ such that $T_1 = ST_2$. If $\text{range } T_1 = \text{range } T_2$, then there exists an invertible operator that's linear and maps from V to V such that $T_1 = T_2S$.

Observe that if ST is invertible, then we must be able multiply it by any S^{-1} or T^{-1} , so that we could find S or T . And if S and T are invertible, their determinants are nonzero, so ST 's determinant is also nonzero, so it's also invertible.

20 If $ST = I$, then ST is invertible, so $(ST)^{-1} = T^{-1}S^{-1} = I$, so multiplying by T on the left gives $S^{-1} = T$, so multiplying by S on the right gives $TS = I$.

If $STU = I$, multiplying by S^{-1} on the left gives $TU = S^{-1}$, then multiplying by U^{-1} on the right gives $T = S^{-1}U^{-1}$, which we know equals $(US)^{-1}$ by problem 1, so $T^{-1} = US$, as desired.

21

- (a) If V isn't finite-dimensional, then S , T , and U might not necessarily have inverses, so there goes the proof. If R , S , and T are all linear maps from V to V itself, since RST is plugging the output of T into S and plugging this into R , then since T is surjective, then S is injective. It's a linear map from $\mathbb{F}$ to V . Since $\dim \mathbb{F} = 1$, then it's isomorphic.
- (b) Observe that if the only solution if $c_1 = \cdots = c_n = 0$ is all 0's, then this means that solutions exist, and its linearly independent ($A_{\cdot,k}$ for all k) so it can also act as a basis, so there will have to exist x_k . And if there exist x_k as solutions to the $c_{\cdot}$, then it's a basis, so it's linearly independent, so $x_1 = \cdots = x_n = 0$ is the only solution if all the $c_{\cdot}$'s equal 0.

7.B The Spectral Theorem

1 Consider the map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^2$. Then if we map $T(a, b, c, d, e)$ to $(a, b, 0, 0, 0)$, then the range has dimension 2, and the null space consists of all quintuplets of the form $(0, 0, c, d, e)$, which has dimension 3. Suppose $S : X \rightarrow Y$ and $T : U \rightarrow V$. Then if we have $x \in X$, which maps to a $y \in Y$, then $y \in \text{null } T$, so $T(y) = T(S(x)) = 0$, so $TS = 0$, so $(ST)^2 = 0$.

2 Since $(v_1, \dots, v_m)$ encompasses all of V , then the map from the vectors, $\mathbb{F}^m$, to V must be surjective. If $v_1, \dots, v_m$ are linearly independent, this means that the null space has degree 0, and consists of only $\{0\}$. Since we know that $\dim \text{null } T + \dim \text{range } T = \dim V$ but $\dim V = 5$ and $\dim \text{null } T > 2$ and $\dim \text{range } T = 4$, which is impossible.

3 Observe that since $\dim \text{range } T \geq \dim V$, but $\dim \text{null } T \geq 1$ because T is not injective, then it is impossible for $\dim \text{range } T + \dim \text{null } T$ to equal $\dim V$. Because T is not surjective, this means that $\dim \text{range } T < \dim W \leq \dim V$ and $\dim \text{null } T$. Then $\dim \text{range } T < \dim W \leq \dim V$. Hence, $\dim \text{null } T + \dim \text{range } T > \dim V$, so we're done.

4 Since $v_1, \dots, v_n$ is linearly independent, then since T is injective, we also know that the null space of T equals 0, so $Tv_1, \dots, Tv_n$ is also linearly independent because we know that T is a linear map, so $Tv + Tw = T(v + w)$. And since its injective, each v maps to a distinct output, so we can look at the inputs instead, which are linearly independent.

5 Since T is a linear map from V to W and $v_1, \dots, v_n$ spans V this also means that $Tv_1, \dots, Tv_n$ spans W . $(S_1 S_2 \cdots S_n)(v) = (S_1(v))(S_2(v)) \cdots (S_n(v))$. So if we have another w such that $(S_1 S_2 \cdots S_n)(w) = (S_1 S_2 \cdots S_n)(v)$, then we must have $S_1(v) = S_1(w)$, etc, so since $S_1, \dots, S_n$ are injective, then $v = w$.

6 We want to show that T is surjective. Obviously, if T is surjective, then U suffices as the input because we really only need one zero, and the rest would be taken care of if we let U be the set consisting of all inputs that don't result in an output of 0, plus one element from the null set, which is $\{0\}$. Observe that the null space consists of points of the form $(5a, a, 7b, b)$, which has dimension 2. Since $\dim \text{null } T + \dim \text{range } T = \dim V = 4$, then $\dim \text{range } T = 2 \dim W$, so T is surjective, as desired.

7 Since $\text{null } T = U$, then since U is 3-dimensional, then $\dim \text{null } T = 3$ as well. $\dim \text{null } T + \dim \text{range } T = \dim V = 8$, so $\dim \text{range } T = 5 = \dim W$, so T is surjective, as desired. So the null space consists of quintuplets of the form $(3a, a, b, b, b)$, which has dimension 2. Since $\dim \text{range } T = \dim V - \dim \text{null } T = 5 - 2 = 3$, then $\dim \text{range } T > \dim W$, which is impossible. Thus there exists values such that

$$Tv = \varphi_1(v)w_1 + \cdots + \varphi_m(v)w_m.$$

8 Consider the map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^2$. Then if we map $T(a, b, c, d, e)$ to $(a, b, 0, 0, 0)$, then the range has dimension 2, and the null space consists of all quintuplets of the form

$(0, 0, c, d, e)$, which has dimension 3. Suppose $S : X \rightarrow Y$ and $T : U \rightarrow V$. Then if we have $x \in X$, which maps to a $y \in Y$, then $y \in \text{null } T$, so $T(y) = T(S(x)) = 0$, so $TS = 0$, so $(ST)^2 = 0$.

9 Since $(v_1, \dots, v_m)$ encompasses all of V , then the map from the vectors, $\mathbb{F}^m$, to V must be surjective. If $v_1, \dots, v_m$ are linearly independent, this means that the null space has degree 0, and consists of only $\{0\}$. Since we know that $\dim \text{null } T + \dim \text{range } T = \dim V$ but $\dim V = 5$ and $\dim \text{null } T > 2$ and $\dim \text{range } T = 4$, which is impossible.

10 Observe that if both T (which is a linear map from U to V) and S which is a linear map from V to W are invertible, then ST , which is a linear map from U to W is invertible, since it's the product of S and T , and thus $ST = T^{-1}S^{-1}$. This doesn't hold because the sum of two noninvertible operators might just be invertible.

11 If there exists an invertible operator $T \in \mathcal{L}(V)$ such that $Tu = Su$ for every $u \in U$, then this must mean that since T is both injective and surjective, then since $Tu = Su$, then so must S . Conversely, if S is injective, then there must exist T such that $Tu = Su$.

12 If $\text{null } T_1 = \text{null } T_2$, this implies that the map from the output of T_2 to the output of T_1 is linear, and it exists, so there exists $S \in \mathcal{L}(W)$ such that $T_1 = ST_2$. If $\text{range } T_1 = \text{range } T_2$, then there exists an invertible operator that's linear and maps from V to V such that $T_1 = T_2S$. Since T is a linear map from V to W and $v_1, \dots, v_n$ spans V this also means that $Tv_1, \dots, Tv_n$ spans W .

13 $(S_1S_2 \cdots S_n)(v) = (S_1(v))(S_2(v)) \cdots (S_n(v))$. So if we have another w such that $(S_1S_2 \cdots S_n)(w) = (S_1S_2 \cdots S_n)(v)$, then we must have $S_1(v) = S_1(w)$, etc, so since $S_1, \dots, S_n$ are injective, then $v = w$.

We want to show that T is surjective. Obviously, if T is surjective, then U suffices as the input because we really only need one zero, and the rest would be taken care of if we let U be the set consisting of all inputs that don't result in an output of 0, plus one element from the null set, which is $\{0\}$.

14 Observe that if ST is invertible, then we must be able multiply it by any S^{-1} or T^{-1} , so that we could find S or T . And if S and T are invertible, their determinants are nonzero, so ST 's determinant is also nonzero, so its also invertible.

If $ST = I$, then ST is invertible, so $(ST)^{-1} = T^{-1}S^{-1} = I$, so multiplying by T on the left gives $S^{-1} = T$, so multiplying by S on the right gives $TS = I$.

15 Since they are saying that the poor lone kid isn't normal, then this implies that they themselves are normal. Specifically, the kid with black hair is normal, and we know 8 out of 9 entries. Let the matrix be

$$T = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & x \end{pmatrix}.$$

Then

$$T^* = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & x \end{pmatrix}.$$

Then

$$TT^* = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & x \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & x \end{pmatrix} = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & x & 1+x^2 \end{pmatrix}$$

and

$$T^*T = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & x \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & x \end{pmatrix} = \begin{pmatrix} 2 & 1 & x \\ 1 & 2 & 1 \\ x & 1 & 1+x^2 \end{pmatrix}.$$

Since T is normal, then $TT^* = T^*T$, so $x = 1$.

7.C Positive Operators and Isometries

1 Let φ_1 be the functional that maps $\begin{pmatrix} a \\ b \end{pmatrix}$ to a and φ_2 be the functional that maps $\begin{pmatrix} a \\ b \end{pmatrix}$ to b . Since T' represents the linear map from $(\mathbb{R}^2)'$ to $(\mathbb{R}^3)'$, then $T'(\varphi_1)$ and $T'(\varphi_2)$ are linear maps from $\mathbb{R}^2$ to $\mathbb{R}$.

2 ψ_1 would be the functional that maps $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ to a , ψ_2 would be the functional that maps $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ to b , and ψ_3 would be the functional that maps $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ to c . Then, since ψ_1 , ψ_2 , and ψ_3 are parts of a basis, so they are (basis)es though not really, so $T'(\varphi_1)$ and $T'(\varphi_2)$ are linear combinations of ψ_1 , ψ_2 , ψ_3 .

3 Well, T is a linear map, and $(Tp)(x)$ is a linear transformation on $p(x)$. Then since $\varphi(p)$ is a functional that maps $\mathcal{P}(\mathbb{R})$ to $\mathbb{R}$. T is a linear map from $\mathcal{P}(\mathbb{R})$ to $\mathcal{P}(\mathbb{R})$, so T' is a linear map from $\mathcal{P}(\mathbb{R})'$ to $\mathcal{P}(\mathbb{R})'$, which is basically a linear map from (the set of functionals from $\mathbb{P}(\mathbb{R})$ to $\mathbb{R}$) to itself. Thus $T'(\varphi) = \varphi(Tp) = \varphi(x^2p(x) + p''(x))$. Since $(x^2p(x) + p''(x))' = 2xp(x) + x^2p'(x) + p'''(x)$, then plugging in $x = 4$ gives $8p(4) + 16p'(4) + p'''(4)$.

4 Well, since T' maps from $\mathcal{L}(W, \mathbb{R})$ to $\mathcal{L}(V, \mathbb{R})$, then if $T' = 0$, it would also imply that T , which maps from V to W equals 0, and vice versa.

5 Since T is an element of $\mathcal{L}(V, W)$ and T' is an element of $\mathcal{L}(W', V')$, and since these two are distinct and don't have any common elements, it suffices to prove that they have an equal number of elements. Since $(v_1, \dots, v_m)$ encompasses all of V , then the map from the vectors, $\mathbb{F}^m$, to V must be surjective. If $v_1, \dots, v_m$ are linearly independent, this means that the null space has degree 0, and consists of only $\{0\}$.

6 Since T is positive, then T has a square root. Say $R^2 = T$, so then $T^k = (R^k)^1$. Since T^k also has a square root, then T^k is positive.

7 Obviously, if $\varphi(u) = 0$ for some u , then u is automatically an element of null φ . Therefore, if for all $u \in U$, $\varphi(u) = 0 \implies u \in \text{null } \varphi$, then $U \subset \text{null } \varphi$.

8 If $U = \{0\}$, then for the only $u \in U$ is $u = 0$. Obviously, for all $\varphi \in V'$, $\varphi(u) = 0$ for any $u \in U$, namely because there's only one value of u and $\varphi(0) = 0$. Therefore, if $U = \{0\}$, then $U^0 = V'$. Conversely, if for all $\varphi \in V'$, $\varphi(u) = 0$ for every $u \in U$, wouldn't this imply that V' is severely limited? So U can only have one value, namely 0.

9 Observe that if $U^0 = 0$, then only the function $\varphi = 0$ satisfies $\varphi(u) = 0$ for all $u \in U$, so $U = V$. And if $U = V$, then for $\varphi(u) = 0$, we can only have $\varphi = 0$, because then the functionals in V would be constants, so they'd have to equal 0.

Consider U^0 , which consists of all $\varphi \in V'$ such that for all $u \in U$, $\varphi(u) = 0$. And we can also consider W^0 , which consists of all $\varphi \in V'$ such that for all $w \in W$, $\varphi(w) = 0$. Note that if $u \in U$, then $u \in W$, so if $\varphi \in W^0$, then we also have $\varphi \in U^0$, so $W^0 \subset U^0$.

10

- (a) If $W^0 \subset U^0$, then this implies that for any $\varphi \in V'$, W^0 consists of the φ such that $\varphi(w) = 0$ for all $w \in W$, and U^0 consists of the φ such that $\varphi(u) = 0$ for all $u \in U$. Since $W^0 \subset U^0$, then if we have any φ such that $\varphi(w) = 0$ for all $w \in W$, then $\varphi(u) = 0$ for all $u \in U$. Thus $U \subset W$, as desired.
- (b) Consider any $x \in U + W$. Automatically, $x \in U$ and $x \in W$. Therefore, if for any $\varphi \in V'$ such that $\varphi \in (U + W)^0$, then for any $x \in U + W \iff x \in U, x \in W$, $\varphi(x) = 0$. Thus, if $\varphi(x) = 0$ for all x , then for the same φ , we also have $\varphi(u) = 0$ for all $u \in U$ and $\varphi(w) = 0$ for all $w \in W$. Hence, $(U + W)^0 \subset U^0 \cap W^0$.

- (c) However, $U^0 \cap W^0$ consists of all $\varphi \in V'$ such that $\varphi(u) = 0$ for all $u \in U$ and $\varphi(w) = 0$ for all $w \in W$. If, for any x , $x \in U$ and $x \in W$, then $x \in U + W$, so this also implies that $\varphi(x) = 0$ for all $x \in U + W$, so $U^0 \cap W^0 \subset (U + W)^0$, so $U^0 \cap W^0 = (U + W)^0$, as desired.
- (d) If $x \in U \cap W$, then $x \in U$ and $x \in W$. Therefore, if $\varphi(x) = 0$ for all $x \in U \cap W$, then $\varphi(x) = 0$ for all $x \in U$, and $\varphi(x) = 0$ for all $x \in W$. Hence, $(U \cap W)^0 \subset U^0 + W^0$. Additionally, if $\varphi \in U^0 + W^0$, then $\varphi(u) = 0$ for all $u \in U$ and $\varphi(w) = 0$ for all $w \in W$, so $\varphi(x) = 0$ if $x \in U \cap W$. Thus $(U \cap W)^0 = U^0 + W^0$.

11 Consider a basis $u_1, \dots, u_m$ of U , and extend it to a basis $u_1, \dots, u_m, \dots, u_n$ of V , so basically $\dim U = m$ and $\dim V = n$. Now let $\varphi_1, \dots, \varphi_m, \dots, \varphi_n$ be the dual basis of V' , so then $\varphi_{m+1}, \dots, \varphi_n$ can be a basis of U^0 . Thus $\dim U^0 = n - m$, so $\dim U + \dim U^0 = \dim V$, as desired.

12 If $\varphi(v) = 0$ for every $\varphi \in U^0$ and we are looking for the v , then since $\varphi \in U^0$, we already know that $\varphi(v) = 0$ for all $v \in V$. Thus the set is U . This is like taking the reverse of the reverse; it goes back to the original. We're looking for the v , then taking the annihilator or something. So this gives $\Gamma = \{v \in V : \varphi(v) = 0 \text{ for every } \varphi \in \Gamma\}^0$.

13 T' is the linear map from $\mathcal{L}(\mathcal{P}_5(\mathbb{R}), \mathbb{R})$ to $\mathcal{L}(\mathcal{P}_5(\mathbb{R}), \mathbb{R})$. Thus, the range of T consists of all p such that $\varphi(p) = 0$. But $\varphi(p) = p(8)$, so it consists of all p such that $p(8) = 0$. Observe that if both T (which is a linear map from U to V) and S which is a linear map from V to W are invertible, then ST , which is a linear map from U to W is invertible, since it's the product of S and T , and thus $ST = T^{-1}S^{-1}$.

14 This doesn't hold because the sum of two noninvertible operators might just be invertible. If there exists an invertible operator $T \in \mathcal{L}(V)$ such that $Tu = Su$ for every $u \in U$, then this must mean that since T is both injective and surjective, then since $Tu = Su$, then so must S . Conversely, if S is injective, then there must exist T such that $Tu = Su$.

7.D Polar Decomposition and Singular Value Decomposition

1 If V is finite-dimensional and $\varphi_1, \dots, \varphi_n$ is a basis of V' . Then there must be a basis of V whose dual basis is $\varphi_1, \dots, \varphi_n$ because V' is the dual space of V . V' is the set of all linear maps from V to $\mathbb{F}$, and since $\varphi_1, \dots, \varphi_n$ are the functionals such that there exists a basis $v_1, \dots, v_n$ of V such that $\varphi_i(v_j) = 1$ if and only if $i = j$, and 0 otherwise.

2 We show that everything is equivalent to T is invertible. Obviously, if the columns of $\mathcal{M}(T)$ are linearly independent in $\mathbb{F}^{n,1}$, then it can form a basis (since its linearly independent, and everything that's linearly independent can form a basis), so they are equivalent. Then, if they span $\mathbb{F}^{n,1}$, T is invertible. Similarly, if the rows of $\mathcal{M}(T)$ are linearly independent, it's equivalent to being a basis of $\mathbb{F}^{1,n}$, so it spans $\mathbb{F}^{1,n}$, and vice-versa. Therefore, these 5 statements are equivalent to each other.

3 We already know that Λ maps from V to V'' , so it suffices to show that this transformation is linear. Since $(\Lambda v)(\varphi) = \varphi(v)$, if we have φ_1 and φ_2 , then

$$(\Lambda v)(\varphi_1 + \varphi_2) = (\varphi_1 + \varphi_2)(v) = \varphi_1(v) + \varphi_2(v) = (\Lambda v)(\varphi_1) + (\Lambda v)(\varphi_2).$$

Furthermore,

$$(\Lambda v)(\lambda \varphi) = (\lambda \varphi)(v) = \lambda \cdot \varphi(v) = \lambda (\Lambda v)(\varphi).$$

Therefore, Λ is a linear map from V to V' .

4 Consider U^0 , which consists of all $\varphi \in V'$ such that for all $u \in U$, $\varphi(u) = 0$. And we can also consider W^0 , which consists of all $\varphi \in V'$ such that for all $w \in W$, $\varphi(w) = 0$. Note that if $u \in U$, then $u \in W$, so if $\varphi \in W^0$, then we also have $\varphi \in U^0$, so $W^0 \subset U^0$.

5 $(\mathcal{P}(\mathbb{R}))'$ is the set of linear maps from $\mathcal{P}(\mathbb{R})$ to $\mathbb{R}$. And $\mathbb{R}^\infty$ is the set of all sequences over $\mathbb{R}$. Since a polynomial is basically a sequence of real numbers (take the coefficients), then via a one-to-one correspondence, we're done.

6 U^0 consists of all $\varphi \in V'$ such that $\varphi(u) = 0$ for all $u \in U$. However, we know that $i(u) = u$ for all $u \in U$. Since i' is a map from V' to U' , then it's a map from φ to U' . The null space of i' is the set of i such that $i(u) = 0$ for all $u \in U$, which is equivalent to U^0 .

7 i' is literally a linear map from V' to U' , so of course its range is U' . We already know that i is a linear map from U to V , so i' is a linear map from V' to U' , so $\tilde{i}'$ is the linear map from $V'/(i') \rightarrow U'$. But, $i' = U^0$, so its the linear map from V'/U^0 to U' , as desired.

8 It suffices to show that the null space of π' , which is the set of linear transformations $\in \mathcal{L}(V/U, \mathbb{R})$ that map to 0, has only one element, which is 0. Since V/U is the set of all affine subsets of V parallel to U , then we are done.

9 Since U^0 is defined as the set of $\varphi \in V'$ such that $\varphi(u) = 0$ for all $u \in U$, then since π' maps to functionals in V' , and $\varphi \in V'$ as well, then $\text{range } \pi' = U^0$, as desired.

10 Since $\text{range } \pi' = U^0$, then π' maps from its domain, $(V/U)'$, to U^0 , and thus its a linear map from $(V/U)'$ to U^0 . And, since its injective, and $\dim(V/U)' = \dim U^0$, so its also surjective. Since its both injective and surjective, then its bijective, so its an isomorphism from $(V/U)'$ onto U^0 .

11 Let $z = a + bi$, so $\bar{z} = a - bi$, so $z + \bar{z} = 2a = 2\text{Re } z$. Let $z = a + bi$, so $\bar{z} = a - bi$, so $z - \bar{z} = 2bi = 2(\text{Im } z)i$. $z\bar{z} = (a + bi)(a - bi) = a^2 + b^2 = (\sqrt{a^2 + b^2})^2 = |z|^2$.

12 Let $w = a + bi$ and $z = c + di$. $w + z = (a + c) + (b + d)i$, so $\overline{w + z} = (a + c) - (b + d)i$. $\bar{w} = a - bi$ and $\bar{z} = c - di$, so $\bar{w} + \bar{z} = (a + c) - (b + d)i = \overline{w + z}$. $wz = (a + bi)(c + di) = (ac - bd) + (bc + da)i$, so $\overline{wz} = (ac - bd) - (bc + da)i$. $\bar{w} \cdot \bar{z} = (a - bi)(c - di) = (ac - bd) - (bc + da)i = \overline{wz}$, as desired.

13 Let $w = a + bi$, $z = c + di$. Then $wz = (a + bi)(c + di) = ac - bd + (bc + da)i$, and

$$\begin{aligned} |wz|^2 &= (ac - bd)^2 + (bc + da)^2 \\ &= (a^2c^2 - 2abcd + b^2d^2) + (b^2c^2 + 2abcd + d^2a^2) \\ &= a^2c^2 + b^2d^2 + b^2c^2 + d^2a^2 \end{aligned}$$

Then

$$|w|^2|z|^2 = (a^2 + b^2)(c^2 + d^2) = a^2c^2 + b^2c^2 + b^2d^2 + a^2d^2 = |wz|^2,$$

so since $|w|^2|z|^2 = |wz|^2$, then $|w||z| = |wz|$.

14 $z = a + bi$, $\bar{z} = a - bi$, $\bar{\bar{z}} = \overline{a - bi} = a + bi = z$. $\text{Re } z = a$ and $\text{Im } z = b$. Both a and b are real numbers. Then $|z| = \sqrt{a^2 + b^2}$, so $|z|^2 = a^2 + b^2$. Since they are real numbers, $b^2 \geq 0$ and $a^2 \geq 0$, so $a^2 \leq |z|^2 \implies |a| \leq |z|$ and $b^2 \leq |z|^2 \implies |b| \leq |z|$. $|z| = \sqrt{a^2 + b^2}$ and $|\bar{z}| = \sqrt{a^2 + (-b)^2} = \sqrt{a^2 + b^2} = |z|$, as desired.

15 For example consider, say, $m = 2$, and the polynomials $p_1(x) = x^2$, $p_2(x) = -x^2$. Then $p_1(x) + p_2(x) = 0$, which has degree 0, so its not closed under addition, so its not a subspace of $\mathcal{P}(\mathbb{F})$. Also, consider the polynomials $p_1(x) = x^2 + x$ and $p_2(x) = -x^2 + x$. Then $p_1(x) + p_2(x) = 2x$, which is an odd degreed polynomial, so its not closed under addition, so its not a subspace of $\mathcal{P}(\mathbb{F})$.

If $m \leq n$, we can go with something like $p = (x - \lambda_1)^{n-m+1}(x - \lambda_2) \cdots (x - \lambda_m)$, which certainly has degree n and has only $\lambda_1, \dots, \lambda_m$ as zeros.

16 We can use the method of undetermined coefficients, but if we let $z_1, \dots, z_{m+1}$ be a basis of $\mathbb{F}$, then we can also let $w_1, \dots, w_{m+1}$ be its dual basis, so there does exist p such that $p(z_j) = w_j$.

If p has m distinct zeros, then this means that each zero occurs exactly once, so p looks something like $p(x) = c(x - \lambda_1) \cdots (x - \lambda_m)$. Then $p'(\lambda_j) \neq 0$ for any $1 \leq j \leq m$, so p and p' have no zeros in common, and the reverse is also true.

Well, if you need it to have real coefficients, then the imaginary roots have to come in pairs, so there are an even number of imaginary roots. However, if we have a polynomial of odd degree, then there must be an odd number of roots, so there has to be at least one root that is real.

17

- (a) Observe that if p is a polynomial, then p' is also a polynomial, and when $x = 3$, Tp is a constant. Now suppose $x \neq 3$. Let $p(x) = k + (x - 3)q(x)$ where q is also a polynomial, so k is the remainder when $p(x)$ is divided by $x - 3$. Then $p(3) = k$, so $p(x) - k = p(x) - p(3) = (x - 3)q(x)$. Hence, $\frac{p(x) - p(3)}{x - 3} = q(x)$, which is also a polynomial. Thus $Tp \in \mathcal{P}(\mathbb{R})$.

Also, observe that T is continuous because $\lim_{x \rightarrow 3} \frac{p(x) - p(3)}{x - 3} = p'(3)$ by the definition of derivative, so T is a linear map. Also, since in Tv , the e_i are in the inner product and the f_i are outside, then it makes sense that T^*v is reversed, so that the f_i are in the inner product and the e_i are outside, which gives the expression. Also, if T is self-adjoint, then the $e_1, \dots, e_n$ and the $f_1, \dots, f_n$ are equal, so $Tv = T^*v$.

- (b) Since T is e_i inside, f_i outside, and T^* is f_i inside, e_i outside, then if we plug T into T^* , then its e_i inside, e_i outside, plus an extra factor resulting from multiplying s_i by itself. Observe that if we let $p(z) = c_0 + c_1z + \cdots + c_nz^n$, then

$$\begin{aligned} p(\bar{z}) &= c_0 + c_1\bar{z} + c_2\bar{z}^2 + \cdots + c_n\bar{z}^n \\ &= c_0 + c_1\bar{z} + c_2\overline{z^2} + \cdots + c_n\overline{z^n} \\ &= \overline{c_0 + c_1z + c_2z^2 + \cdots + c_nz^n} \\ &= \overline{p(z)}. \end{aligned}$$

Thus $\overline{p(\bar{z})} = p(z)$, so $q(z) = p(z)^2$, so it has real coefficients.

- (c) We know that if we can determine the value of p with degree m for $m + 1$ values of x , then we can determine p . Let $p = c_0 + c_1x + \cdots + c_mx^m$, so we know that

$$\begin{aligned} p(x_0) &= c_0 + c_1x_0 + \cdots + c_mx_0^m \\ p(x_1) &= c_0 + c_1x_1 + \cdots + c_mx_1^m \\ &\vdots \\ p(x_m) &= c_0 + c_1x_m + \cdots + c_mx_m^m. \end{aligned}$$

Since we have $m + 1$ unknowns, which are $c_0, \dots, c_m$, and $m + 1$ equations, then we can solve for the c_i . Since all of the x_i are real numbers, and all of the $p(x_i)$ are real numbers, then all of the c_i are also real numbers.

- (d) Observe that $(\lambda(x_1, x_2), (y_1, y_2)) = |\lambda x_1 y_1| + |\lambda x_2 y_2|$. However, if λ is negative, and x_1, x_2, y_1, y_2 are positive, then this is obviously not equal to $\lambda(|x_1 y_1| + |x_2 y_2|)$. This function doesn't satisfy definiteness, because if we have the functions $(0, 1, 0)$ and $(0, 1, 0)$, we'll get an output of 0, even though neither input is 0.

We have

$$\begin{aligned}
 \langle u + v, u - v \rangle &= \langle u, u - v \rangle + \langle v, u - v \rangle \\
 &= \langle u, u \rangle - \langle u, v \rangle + \langle v, u \rangle - \langle v, v \rangle \\
 &= \langle u, u \rangle - \langle v, v \rangle \\
 &= \|u\|^2 - \|v\|^2.
 \end{aligned}$$

If $\|u\| = \|v\|$, then $\langle u + v, u - v \rangle = \|u\|^2 - \|v\|^2 = 0$, so $u + v$ and $u - v$ are orthogonal.

18

- (a) If u and v are two adjacent sides of a rhombus, then the diagonals are $u + v$ and $u - v$ in some order. It's a rhombus, so $\|u\| = \|v\|$, so $u + v$ and $u - v$ are orthogonal by part (b), so the diagonals are perpendicular as desired.
- (b) $T - \lambda I$ being not invertible is equivalent to λ being an eigenvalue, so λ not being an eigenvalue is equivalent to $T - \lambda I$ being invertible. $\sqrt{2}$ is obviously not an eigenvalue because if $\|Tv\| \leq \|v\|$, then $\|Tv\| < \sqrt{2}\|v\|$, so $Tv < \sqrt{2}v$, so we're done.

19 Observe that if $\langle u, v \rangle = 0$, then u and v are orthogonal, so no matter what value a is, av will be orthogonal to u . $u + av$ is the hypotenuse of a right triangle with legs u and av , so $\|u + av\| \geq \|u\|$ by the Pythagorean Theorem.

We prove that if $\|u\| = \|v\|$, then $\|au + bv\| = \|bu + av\|$. Observe that $au + bv$ is the diagonal of a parallelogram with sides au and bv , and $bu + av$ is the diagonal of a parallelogram with sides bu and av . Since $\|u\| = \|v\|$, the parallelogram with sides bu and av is just the reflection of the parallelogram with sides au and bv over the angle bisector of u and v . Hence $\|au + bv\| = \|bu + av\|$.

Also, if $\|au + bv\| = \|bu + av\|$ for all $a, b \in \mathbb{R}$, then letting $a = 1$ and $b = 0$ gives $\|u\| = \|v\|$.

20 Suppose for the sake of contradiction that $\|u\| \leq \|u + av\|$ for all $a \in \mathbb{F}$, and u and v are not orthogonal. Now we will find a value of $a \in \mathbb{F}$ such that $\|u\| > \|u + av\|$. First, draw the vectors u and v , starting at the origin. Make sure their starting points are the origin. Then draw a circle at the origin with radius $\|u\|$, so the head of u is on the circle. Then draw a line through the head of u parallel to v , so that it intersects the circle at some point. Note that if u and v were orthogonal, then it would be tangent. Anyway, if it intersects the circle at points, then if you connect the origin to a point on this line, it will be $u + av$ for some value of a , but its norm would be less than $\|u\|$, as desired.

Chapter 8: Operators on Complex Vector Spaces

Linear Algebra Done Right

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8.A Generalized Eigenvectors and Nilpotent Operators

1 Consider the map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^2$. Then if we map $T(a, b, c, d, e)$ to $(a, b, 0, 0, 0)$, then the range has dimension 2, and the null space consists of all quintuplets of the form $(0, 0, c, d, e)$, which has dimension 3. Suppose $S : X \rightarrow Y$ and $T : U \rightarrow V$. Then if we have $x \in X$, which maps to a $y \in Y$, then $y \in \text{null } T$, so $T(y) = T(S(x)) = 0$, so $TS = 0$, so $(ST)^2 = 0$.

2 Since $(v_1, \dots, v_m)$ encompasses all of V , then the map from the vectors, $\mathbb{F}^m$, to V must be surjective. If $v_1, \dots, v_m$ are linearly independent, this means that the null space has degree 0, and consists of only $\{0\}$.

3 Since we know that $\dim \text{null } T + \dim \text{range } T = \dim V$ but $\dim V = 5$ and $\dim \text{null } T > 2$ and $\dim \text{range } T = 4$, which is impossible. How about something like $T : (a, b, c, d) \rightarrow (a, b, 0, 0)$. Then the range has dimension 2 but the null space also has dimension 2. Because we know that $\dim V = \dim \text{range } T + \dim \text{null } T$, but $\dim V = 5$, which is odd, so we can't split it evenly.

4 Observe that since $\dim \text{range } T \geq \dim V$, but $\dim \text{null } T \geq 1$ because T is not injective, then it is impossible for $\dim \text{range } T + \dim \text{null } T$ to equal $\dim V$. Because T is not surjective, this means that $\dim \text{range } T < \dim W \leq \dim V$ and $\dim \text{null } T$. Then $\dim \text{range } T < \dim W \leq \dim V$. Hence, $\dim \text{null } T + \dim \text{range } T > \dim V$, so we're done.

5 Since $v_1, \dots, v_n$ is linearly independent, then since T is injective, we also know that the null space of T equals 0, so $Tv_1, \dots, Tv_n$ is also linearly independent because we know that T is a linear map, so $Tv + Tw = T(v + w)$. And since it's injective, each v maps to a distinct output, so we can look at the inputs instead, which are linearly independent. Since T is a linear map from V to W and $v_1, \dots, v_n$ spans V this also means that $Tv_1, \dots, Tv_n$ spans W .

6 $(S_1 S_2 \cdots S_n)(v) = (S_1(v))(S_2(v)) \cdots (S_n(v))$. So if we have another w such that $(S_1 S_2 \cdots S_n)(w) = (S_1 S_2 \cdots S_n)(v)$, then we must have $S_1(v) = S_1(w)$, etc, so since $S_1, \dots, S_n$ are injective, then $v = w$.

7 We want to show that T is surjective. Obviously, if T is surjective, then U suffices as the input because we really only need one zero, and the rest would be taken care of if we let U be the set consisting of all inputs that don't result in an output of 0, plus one element from the null set, which is $\{0\}$.

8 Observe that the null space consists of points of the form $(5a, a, 7b, b)$, which has dimension 2. Since $\dim \text{null } T + \dim \text{range } T = \dim V = 4$, then $\dim \text{range } T = 2$, so T is surjective, as desired.

9 Since $\text{null } T = U$, then since U is 3-dimensional, then $\dim \text{null } T = 3$ as well. $\dim \text{null } T + \dim \text{range } T = \dim V = 8$, so $\dim \text{range } T = 5 = \dim W$, so T is surjective, as desired.

10 So the null space consists of quintuplets of the form $(3a, a, b, b, b)$, which has dimension 2. Since $\dim \text{range } T = \dim V - \dim \text{null } T = 5 - 2 = 3$, then $\dim \text{range } T > \dim W$, which is impossible.

11 Thus there exists values such that

$$Tv = \varphi_1(v)w_1 + \cdots + \varphi_m(v)w_m.$$

Consider the map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^2$. Then if we map $T(a, b, c, d, e)$ to $(a, b, 0, 0, 0)$, then the range has dimension 2, and the null space consists of all quintuplets of the form $(0, 0, c, d, e)$, which has dimension 3.

12 Suppose $S : X \rightarrow Y$ and $T : U \rightarrow V$. Then if we have $x \in X$, which maps to a $y \in Y$, then $y \in \text{null } T$, so $T(y) = T(S(x)) = 0$, so $TS = 0$, so $(ST)^2 = 0$.

13 Since $(v_1, \dots, v_m)$ encompasses all of V , then the map from the vectors, $\mathbb{F}^m$, to V must be surjective. If $v_1, \dots, v_m$ are linearly independent, this means that the null space has degree 0, and consists of only $\{0\}$.

14 Since we know that $\dim \text{null } T + \dim \text{range } T = \dim V$ but $\dim V = 5$ and $\dim \text{null } T > 2$ and $\dim \text{range } T = 4$, which is impossible.

15 How about something like $T : (a, b, c, d) \rightarrow (a, b, 0, 0)$. Then the range has dimension 2 but the null space also has dimension 2.

16 Because we know that $\dim V = \dim \text{range } T + \dim \text{null } T$, but $\dim V = 5$, which is odd, so we can't split it evenly.

17 Observe that since $\dim \text{range } T \geq \dim V$, but $\dim \text{null } T \geq 1$ because T is not injective, then it is impossible for $\dim \text{range } T + \dim \text{null } T$ to equal $\dim V$, as desired.

18 Because T is not surjective, this means that $\dim \text{range } T < \dim W \leq \dim V$ and $\dim \text{null } T$. Then $\dim \text{range } T < \dim W \leq \dim V$. Hence, $\dim \text{null } T + \dim \text{range } T > \dim V$, so we're done.

19 Since $v_1, \dots, v_n$ is linearly independent, then since T is injective, we also know that the null space of T equals 0, so $Tv_1, \dots, Tv_n$ is also linearly independent because we know that T is a linear map, so $Tv + Tw = T(v + w)$. And since it's injective, each v maps to a distinct output, so we can look at the inputs instead, which are linearly independent.

20 Since T is a linear map from V to W and $v_1, \dots, v_n$ spans V this also means that $Tv_1, \dots, Tv_n$ spans W . $(S_1 S_2 \cdots S_n)(v) = (S_1(v))(S_2(v)) \cdots (S_n(v))$. So if we have another w such that $(S_1 S_2 \cdots S_n)(w) = (S_1 S_2 \cdots S_n)(v)$, then we must have $S_1(v) = S_1(w)$, etc, so since $S_1, \dots, S_n$ are injective, then $v = w$.

21 We want to show that T is surjective. Obviously, if T is surjective, then U suffices as the input because we really only need one zero, and the rest would be taken care of if we let U be the set consisting of all inputs that don't result in an output of 0, plus one element from the null set, which is $\{0\}$.

Observe that the null space consists of points of the form $(5a, a, 7b, b)$, which has dimension 2. Since $\dim \text{null } T + \dim \text{range } T = \dim V = 4$, then $\dim \text{range } T = 2 = \dim W$, so T is surjective, as desired.

8.B Decomposition of an Operator

1 Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. There exists a basis $w_1, \dots, w_n$ of W such that all the entries in the first column of $\mathcal{M}(T)$ (with respect to

the bases $v_1, \dots, v_m$ and $w_1, \dots, w_n$) are 0 except for possibly T as a linear map from V to W such that there is nothing except possibly, a 0 in the first row and first column. If we have the matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

In this case, we have a matrix with a 1's along the diagonal, meaning places where the number of rows and number of columns are equal, and 0 everywhere else.

2 We know that $\mathcal{M}(T)$ is defined as the set of numbers $a_{i,j}$ for $1 \leq i \leq m$, $1 \leq j \leq n$ such that for all j ,

$$w_j = a_{1,j}v_1 + a_{2,j}v_2 + \dots + a_{m,j}v_m.$$

So, if we add w_{s_j} to w_{t_j} (these are the respective bases of S and T), and let a and b denote the values of the entries in the matrices, we get

$$\mathcal{M}(S+T) = (a_{1,i} + b_{1,i})v_1 + \dots + (a_{m,i} + b_{m,i})v_m$$

and

$$\begin{aligned} \mathcal{M}(S) + \mathcal{M}(T) &= [a_{1,i}v_1 + \dots + a_{m,i}v_n] \\ &\quad + [b_{1,i}v_1 + \dots + b_{m,i}v_n] \\ &= (a_{1,i} + b_{1,i})v_1 + \dots + (a_{m,i} + b_{m,i})v_m \end{aligned}$$

as desired.

3 If we have

$$Tv_i = a_{1,i}v_1 + \dots + a_{m,i}v_i,$$

then if we multiply T times λ , we get

$$\lambda a_{1,i}v_1 + \dots + \lambda a_{m,i}v_i = a_{1,i}(\lambda v_1) + \dots + a_{m,i}(\lambda v_i)$$

as desired.

4 Observe that we wish to prove that when we multiply a matrix by a vector, we get a combination of all of the previous vectors and put them together. This is clear because we already have a collection of vectors to start with. A single value times matrix put together gives a vector of values times matrix.

5 Consider $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and $C = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$. Then

$$AC = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$$

and

$$CA = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$$

as desired.

6 Observe that if $(AB)C$ makes sense, then if A is $m \times n$, then B must be $n \times p$ for some p , so AB is $m \times p$, so C is $p \times q$ for some q . Thus BC is $n \times q$, so $A(BC)$ is $m \times q$, which is defined.

7 Observe that if both T (which is a linear map from U to V) and S which is a linear map from V to W are invertible, then ST , which is a linear map from U to W is invertible, since it's the product of S and T , and thus $ST = T^{-1}S^{-1}$. This doesn't hold because the sum of two noninvertible operators might just be invertible.

8 If there exists an invertible operator $T \in \mathcal{L}(V)$ such that $Tu = Su$ for every $u \in U$, then this must mean that since T is both injective and surjective, then since $Tu = Su$, then so must S . Conversely, if S is injective, then there must exist T such that $Tu = Su$. If $\text{null } T_1 = \text{null } T_2$, this implies that the map from the output of T_2 to the output of T_1 is linear, and it exists, so there exists $S \in \mathcal{L}(W)$ such that $T_1 = ST_2$.

9 We can treat these block diagonal matrices as we would any other matrix. We know that

$$AB = \begin{pmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_m \end{pmatrix} \begin{pmatrix} B_1 & & 0 \\ & \ddots & \\ 0 & & B_m \end{pmatrix} = \begin{pmatrix} A_1 B_1 & & 0 \\ & \ddots & \\ 0 & & A_m B_m \end{pmatrix},$$

and since A_j has the same size as B_j for $j = 1, \dots, m$, then AB is clearly defined, so we are done.

10 Observe that if ST is invertible, then we must be able multiply it by any S^{-1} or T^{-1} , so that we could find S or T . And if S and T are invertible, their determinants are nonzero, so ST 's determinant is also nonzero, so its also invertible. If $ST = I$, then ST is invertible, so $(ST)^{-1} = T^{-1}S^{-1} = I$, so multiplying by T on the left gives $S^{-1} = T$, so multiplying by S on the right gives $TS = I$.

11 If $STU = I$, multiplying by S^{-1} on the left gives $TU = S^{-1}$, then multiplying by U^{-1} on the right gives $T = S^{-1}U^{-1}$, which we know equals $(US)^{-1}$ by problem 1, so $T^{-1} = US$, as desired. If V isn't finite-dimensional, then S , T , and U might not necessarily have inverses, so there goes the proof. If R , S , and T are all linear maps from V to V itself, since RST is plugging the output of T into S and plugging this into R , then since T is surjective, then S is injective.

8.C Characteristic and Minimal Polynomials

1 Since we know that the eigenvalues of T are 3, 5, and 8, and that T is a linear map on $\mathbb{C}^4$, which has degree 4, then one of the eigenvalues must have multiplicity 2. Thus $(T-3I)^2(T-5I)(T-8I) = 0$ or $(T-3I)(T-5I)^2(T-8I) = 0$ or $(T-3I)(T-5I)(T-8I)^2 = 0$, so $(T-3I)^2(T-5I)^2(T-8I)^2 = 0$.

2 Since both 5 and 6 are eigenvalues, then we know that they each have multiplicity at most $n-1$ and at least 1. Thus at the extremes, we have $(T-5I)^{n-1}(T-6I) = 0$ or $(T-5I)(T-6I)^{n-1} = 0$. So $(T-5I)^{n-1}(T-6I)^{n-1} = 0$.

3 Take

$$\mathcal{M}(T) = \begin{pmatrix} 7 & 0 & 0 & 0 \\ 0 & 7 & 0 & 0 \\ 0 & 0 & 8 & 0 \\ 0 & 0 & 0 & 8 \end{pmatrix}.$$

Or $T(x_1, x_2, x_3, x_4) = (7x_1, 7x_2, 8x_3, 8x_4)$.

4 Consider

$$\mathcal{M}(T) = \begin{pmatrix} 1 & & & \\ 0 & 5 & 1 & \\ 0 & 0 & 5 & \\ 0 & 0 & 0 & 5 \end{pmatrix}.$$

5 We can take

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

Clearly 0, 1, and 3 are the eigenvalues. Since $T(T - I)(T - 3I) \neq 0$, then the minimal polynomial is $z(z - 1)^2(z - 3)$, and obviously that's the characteristic polynomial as well.

6 It's a linear map from $\mathbb{F}$ to V . Since $\dim \mathbb{F} = 1$, then it's isomorphic. Since $\deg Tp \leq \deg p$ for all p , this must imply that T is a bijection (since it's also injective), so it must be surjective.

7 Since it's surjective, then $\deg Tp = \deg p$ otherwise it wouldn't be surjective. Observe that if the only solution if $c_1 = \dots = c_n = 0$ is all 0's, then this means that solutions exist, and it's linearly independent ($A_{\cdot,k}$ for all k) so it can also act as a basis, so there will have to exist x_k . And if there exist x_k as solutions to the c ., then it's a basis, so it's linearly independent, so $x_1 = \dots = x_n = 0$ is the only solution if all the c 's equal 0.

8 Consider the map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^2$. Then if we map $T(a, b, c, d, e)$ to $(a, b, 0, 0, 0)$, then the range has dimension 2, and the null space consists of all quintuplets of the form $(0, 0, c, d, e)$, which has dimension 3.

9 Suppose $S : X \rightarrow Y$ and $T : U \rightarrow V$. Then if we have $x \in X$, which maps to a $y \in Y$, then $y \in \text{null } T$, so $T(y) = T(S(x)) = 0$, so $TS = 0$, so $(ST)^2 = 0$. Since $(v_1, \dots, v_m)$ encompasses all of V , then the map from the vectors, $\mathbb{F}^m$, to V must be surjective. If $v_1, \dots, v_m$ are linearly independent, this means that the null space has degree 0, and consists of only $\{0\}$.

10 Since we know that $\dim \text{null } T + \dim \text{range } T = \dim V$ but $\dim V = 5$ and $\dim \text{null } T > 2$ and $\dim \text{range } T = 4$, which is impossible. How about something like $T : (a, b, c, d) \rightarrow (a, b, 0, 0)$. Then the range has dimension 2 but the null space also has dimension 2. Because we know that $\dim V = \dim \text{range } T + \dim \text{null } T$, but $\dim V = 5$, which is odd, so we can't split it evenly.

11 Observe that since $\dim \text{range } T \geq \dim V$, but $\dim \text{null } T \geq 1$ because T is not injective, then it is impossible for $\dim \text{range } T + \dim \text{null } T$ to equal $\dim V$, as desired. Because T is not surjective, this means that $\dim \text{range } T < \dim W \leq \dim V$ and $\dim \text{null } T$. Then $\dim \text{range } T < \dim W \leq \dim V$. Hence, $\dim \text{null } T + \dim \text{range } T > \dim V$, so we're done.

12 Since $v_1, \dots, v_n$ is linearly independent, then since T is injective, we also know that the null space of T equals 0, so $Tv_1, \dots, Tv_n$ is also linearly independent because we know that T is a linear map, so $Tv + Tw = T(v + w)$. And since it's injective, each v maps to a distinct output, so we can look at the inputs instead, which are linearly independent.

13 Since T is a linear map from V to W and $v_1, \dots, v_n$ spans V this also means that $Tv_1, \dots, Tv_n$ spans W . $(S_1 S_2 \dots S_n)(v) = (S_1(v))(S_2(v)) \dots (S_n(v))$. So if we have another w such that $(S_1 S_2 \dots S_n)(w) = (S_1 S_2 \dots S_n)(v)$, then we must have $S_1(v) = S_1(w)$, etc, so since $S_1, \dots, S_n$ are injective, then $v = w$.

14 We want to show that T is surjective. Obviously, if T is surjective, then U suffices as the input because we really only need one zero, and the rest would be taken care of if we let U be the set consisting of all inputs that don't result in an output of 0, plus one element from the null set, which is $\{0\}$.

15

- (a) Observe that the null space consists of points of the form $(5a, a, 7b, b)$, which has dimension 2. Since $\dim \text{null } T + \dim \text{range } T = \dim V = 4$, then $\dim \text{range } T = 2 \dim W$, so T is surjective, as desired. Since $\text{null } T = U$, then since U is 3-dimensional, then $\dim \text{null } T = 3$ as well. $\dim \text{null } T + \dim \text{range } T = \dim V = 8$, so $\dim \text{range } T = 5 = \dim W$, so T is surjective, as desired.
- (b) So the null space consists of quintuplets of the form $(3a, a, b, b, b)$, which has dimension 2. Since $\dim \text{range } T = \dim V - \dim \text{null } T = 5 - 2 = 3$, then

$\dim \text{range } T > \dim W$, which is impossible. Thus there exists values such that

$$Tv = \varphi_1(v)w_1 + \cdots + \varphi_m(v)w_m.$$

16 Consider the map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^2$. Then if we map $T(a, b, c, d, e)$ to $(a, b, 0, 0, 0)$, then the range has dimension 2, and the null space consists of all quintuplets of the form $(0, 0, c, d, e)$, which has dimension 3. Suppose $S : X \rightarrow Y$ and $T : U \rightarrow V$. Then if we have $x \in X$, which maps to a $y \in Y$, then $y \in \text{null } T$, so $T(y) = T(S(x)) = 0$, so $TS = 0$, so $(ST)^2 = 0$.

17 Since $(v_1, \dots, v_m)$ encompasses all of V , then the map from the vectors, $\mathbb{F}^m$, to V must be surjective. If $v_1, \dots, v_m$ are linearly independent, this means that the null space has degree 0, and consists of only $\{0\}$. Since we know that $\dim \text{null } T + \dim \text{range } T = \dim V$ but $\dim V = 5$ and $\dim \text{null } T > 2$ and $\dim \text{range } T = 4$, which is impossible.

18 Observe that if both T (which is a linear map from U to V) and S which is a linear map from V to W are invertible, then ST , which is a linear map from U to W is invertible, since it's the product of S and T , and thus $ST = T^{-1}S^{-1}$. This doesn't hold because the sum of two noninvertible operators might just be invertible.

19 If there exists an invertible operator $T \in \mathcal{L}(V)$ such that $Tu = Su$ for every $u \in U$, then this must mean that since T is both injective and surjective, then since $Tu = Su$, then so must S . Conversely, if S is injective, then there must exist T such that $Tu = Su$. If $\text{null } T_1 = \text{null } T_2$, this implies that the map from the output of T_2 to the output of T_1 is linear, and it exists, so there exists $S \in \mathcal{L}(W)$ such that $T_1 = ST_2$.

20 If $\text{range } T_1 = \text{range } T_2$, then there exists an invertible operator that's linear and maps from V to V such that $T_1 = T_2S$. Since T is a linear map from V to W and $v_1, \dots, v_n$ spans V this also means that $Tv_1, \dots, Tv_n$ spans W . $(S_1S_2 \cdots S_n)(v) = (S_1(v))(S_2(v)) \cdots (S_n(v))$. So if we have another w such that

$$(S_1S_2 \cdots S_n)(w) = (S_1S_2 \cdots S_n)(v),$$

then we must have $S_1(v) = S_1(w)$, etc, so since $S_1, \dots, S_n$ are injective, then $v = w$.

8.D Jordan Form

1 We want to show that T is surjective. Obviously, if T is surjective, then U suffices as the input because we really only need one zero, and the rest would be taken care of if we let U be the set consisting of all inputs that don't result in an output of 0, plus one element from the null set, which is $\{0\}$.

2 Observe that if ST is invertible, then we must be able multiply it by any S^{-1} or T^{-1} , so that we could find S or T . And if S and T are invertible, their determinants are nonzero, so ST 's determinant is also nonzero, so its also invertible.

3 If $ST = I$, then ST is invertible, so $(ST)^{-1} = T^{-1}S^{-1} = I$, so multiplying by T on the left gives $S^{-1} = T$, so multiplying by S on the right gives $TS = I$. If $STU = I$, multiplying by S^{-1} on the left gives $TU = S^{-1}$, then multiplying by U^{-1} on the right gives $T = S^{-1}U^{-1}$, which we know equals $(US)^{-1}$ by problem 1, so $T^{-1} = US$, as desired.

4 Let φ_1 be the functional that maps $\begin{pmatrix} a \\ b \end{pmatrix}$ to a and φ_2 be the functional that maps $\begin{pmatrix} a \\ b \end{pmatrix}$ to b . Since T' represents the linear map from $(\mathbb{R}^2)'$ to $(\mathbb{R}^3)'$, then $T'(\varphi_1)$ and $T'(\varphi_2)$ are linear maps from $\mathbb{R}^2$ to $\mathbb{R}$.

5 ψ_1 would be the functional that maps $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ to a , ψ_2 would be the functional that

maps $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ to b , and ψ_3 would be the functional that maps $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ to c . Then, since ψ_1 , ψ_2 , and ψ_3 are parts of a basis, so they are (basis)es though not really, so $T'(\varphi_1)$ and $T'(\varphi_2)$ are linear combinations of ψ_1 , ψ_2 , ψ_3 .

6 Well, T is a linear map, and $(Tp)(x)$ is a linear transformation on $p(x)$. Then since $\varphi(p)$ is a functional that maps $\mathcal{P}(\mathbb{R})$ to $\mathbb{R}$. T is a linear map from $\mathcal{P}(\mathbb{R})$ to $\mathcal{P}(\mathbb{R})$, so T' is a linear map from $\mathcal{P}(\mathbb{R})'$ to $\mathcal{P}(\mathbb{R})'$, which is basically a linear map from (the set of functionals from $\mathbb{P}(\mathbb{R})$ to $\mathbb{R}$) to itself. Thus $T'(\varphi) = \varphi(Tp) = \varphi(x^2p(x) + p''(x))$. Since $(x^2p(x) + p''(x))' = 2xp(x) + x^2p'(x) + p'''(x)$, then plugging in $x = 4$ gives $8p(4) + 16p'(4) + p'''(4)$.

7 Since T is an element of $\mathcal{L}(V, W)$ and T' is an element of $\mathcal{L}(W', V')$, and since these two are distinct and don't have any common elements, it suffices to prove that they have an equal number of elements. Since $(v_1, \dots, v_m)$ encompasses all of V , then the map from the vectors, $\mathbb{F}^m$, to V must be surjective. If $v_1, \dots, v_m$ are linearly independent, this means that the null space has degree 0, and consists of only $\{0\}$. Since T' maps from $\mathcal{L}(W, \mathbb{R})$ to $\mathcal{L}(V, \mathbb{R})$, then if $T' = 0$, it would also imply that T , which maps from V to W equals 0, and vice versa.

8 Obviously, if $\varphi(u) = 0$ for some u , then u is automatically an element of null φ . Therefore, if for all $u \in U$, $\varphi(u) = 0 \implies u \in \text{null } \varphi$, then $U \subset \text{null } \varphi$. If $U = \{0\}$, then for the only $u \in U$ is $u = 0$. Obviously, for all $\varphi \in V'$, $\varphi(u) = 0$ for any $u \in U$, namely because there's only one value of u and $\varphi(0) = 0$. Therefore, if $U = \{0\}$, then $U^0 = V'$. Conversely, if for all $\varphi \in V'$, $\varphi(u) = 0$ for every $u \in U$, wouldn't this imply that V' is severely limited? So U can only have one value, namely 0.

Chapter 9: Operators on Real Vector Spaces

Linear Algebra Done Right

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1 To prove that $\mathcal{L}(V, W)$ is a vector space, we need to prove that addition and multiplication hold for the set of linear maps. So basically, the set of linear maps is a linear map. Hehe. Anyway consider two linear maps, L_1 and L_2 . Then note that we need to prove that $L_1 + L_2$ and λL_1 are also linear maps. If we let $S = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and

$T = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, then $ST = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ and $TS = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$. Clearly $ST \neq TS$.

2 To prove that $L_1 + L_2$ is a linear map, consider two vectors v_1 and v_2 . We want to show that $(L_1 + L_2)(v_1 + v_2) = (L_1 + L_2)v_1 + (L_1 + L_2)v_2$.

$$\begin{aligned} (L_1 + L_2)(v_1 + v_2) &= L_1(v_1 + v_2) + L_2(v_1 + v_2) \\ &= L_1v_1 + L_1v_2 + L_2v_1 + L_2v_2 \\ &= (L_1v_1 + L_2v_1) + (L_1v_2 + L_2v_2) \\ &= (L_1 + L_2)v_1 + (L_1 + L_2)v_2, \end{aligned}$$

since L_1 and L_2 are linear maps. Also,

$$(L_1 + L_2)(\lambda v_1) = L_1(\lambda v_1) + L_2(\lambda v_1) = \lambda L_1v_1 + \lambda L_2v_1 = \lambda(L_1v_1 + L_2v_1) = \lambda(L_1 + L_2)(v_1)$$

so $L_1 + L_2$ is a linear map, as desired.

3 We now show that λL is a linear map.

$$(\lambda L)(v_1 + v_2) = \lambda(L(v_1 + v_2)) = \lambda(Lv_1 + Lv_2) = (\lambda L)v_1 + (\lambda L)v_2$$

so it satisfies summation. Now

$$(\lambda_1 L)(\lambda_2 v) = \lambda_1(L(\lambda_2 v)) = \lambda_1(\lambda_2(Lv)) = \lambda_1\lambda_2(Lv) = \lambda_2((\lambda_1 L)v),$$

as desired. Because L is a linear map itself.

4 Consider the map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^2$. Then if we map $T(a, b, c, d, e)$ to $(a, b, 0, 0, 0)$, then the range has dimension 2, and the null space consists of all quintuplets of the form $(0, 0, c, d, e)$, which has dimension 3. Suppose $S : X \rightarrow Y$ and $T : U \rightarrow V$. Then if we have $x \in X$, which maps to a $y \in Y$, then $y \in \text{null } T$, so $T(y) = T(S(x)) = 0$, so $TS = 0$, so $(ST)^2 = 0$.

5 Since $(v_1, \dots, v_m)$ encompasses all of V , then the map from the vectors, $\mathbb{F}^m$, to V must be surjective. If $v_1, \dots, v_m$ are linearly independent, this means that the null space has degree 0, and consists of only $\{0\}$. Since we know that $\dim \text{null } T + \dim \text{range } T = \dim V$ but $\dim V = 5$ and $\dim \text{null } T > 2$ and $\dim \text{range } T = 4$, which is impossible.

6 How about something like $T : (a, b, c, d) \rightarrow (a, b, 0, 0)$. Then the range has dimension 2 but the null space also has dimension 2. Because we know that $\dim V = \dim \text{range } T +$

$\dim \text{null } T$, but $\dim V = 5$, which is odd, so we can't split it evenly. Observe that since $\dim \text{range } T \geq \dim V$, but $\dim \text{null } T \geq 1$ because T is not injective, then it is impossible for $\dim \text{range } T + \dim \text{null } T$ to equal $\dim V$, as desired.

7 Because T is not surjective, this means that $\dim \text{range } T < \dim W \leq \dim V$ and $\dim \text{null } T > 0$. Then $\dim \text{range } T + \dim \text{null } T < \dim W + \dim \text{null } T \leq \dim V$. Hence, $\dim \text{range } T + \dim \text{null } T < \dim V$, so we're done. Since $v_1, \dots, v_n$ is linearly independent, then since T is injective, we also know that the null space of T equals $\{0\}$, so $Tv_1, \dots, Tv_n$ is also linearly independent because we know that T is a linear map, so $Tv + Tw = T(v + w)$. And since it's injective, each v maps to a distinct output, so we can look at the inputs instead, which are linearly independent.

8 Since T is a linear map from V to W and $v_1, \dots, v_n$ spans V this also means that $Tv_1, \dots, Tv_n$ spans W . $(S_1 S_2 \cdots S_n)(v) = (S_1(v))(S_2(v)) \cdots (S_n(v))$. So if we have another w such that $(S_1 S_2 \cdots S_n)(w) = (S_1 S_2 \cdots S_n)(v)$, then we must have $S_1(v) = S_1(w)$, etc, so since $S_1, \dots, S_n$ are injective, then $v = w$.

9 We want to show that T is surjective. Obviously, if T is surjective, then U suffices as the input because we really only need one zero, and the rest would be taken care of if we let U be the set consisting of all inputs that don't result in an output of 0, plus one element from the null set, which is $\{0\}$.

10 Observe that the null space consists of points of the form $(5a, a, 7b, b)$, which has dimension 2. Since $\dim \text{null } T + \dim \text{range } T = \dim V = 4$, then $\dim \text{range } T = 2$ and $\dim W = 2$, so T is surjective, as desired. Since $\text{null } T = U$, then since U is 3-dimensional, then $\dim \text{null } T = 3$ as well. $\dim \text{null } T + \dim \text{range } T = \dim V = 8$, so $\dim \text{range } T = 5 = \dim W$, so T is surjective, as desired.

11 So the null space consists of quintuplets of the form $(3a, a, b, b, b)$, which has dimension 2. Since $\dim \text{range } T = \dim V - \dim \text{null } T = 5 - 2 = 3$, then $\dim \text{range } T < \dim W$, which is impossible. Thus there exists values such that

$$Tv = \varphi_1(v)w_1 + \cdots + \varphi_m(v)w_m.$$

Consider the map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^2$. Then if we map $T(a, b, c, d, e)$ to $(a, b, 0, 0, 0)$, then the range has dimension 2, and the null space consists of all quintuplets of the form $(0, 0, c, d, e)$, which has dimension 3.

12 Suppose $S : X \rightarrow Y$ and $T : U \rightarrow V$. Then if we have $x \in X$, which maps to a $y \in Y$, then $y \in \text{null } T$, so $T(y) = T(S(x)) = 0$, so $TS = 0$, so $(ST)^2 = 0$. Since $(v_1, \dots, v_m)$ encompasses all of V , then the map from the vectors, $\mathbb{F}^m$, to V must be surjective. If $v_1, \dots, v_m$ are linearly independent, this means that the null space has degree 0, and consists of only $\{0\}$.

13 Since we know that $\dim \text{null } T + \dim \text{range } T = \dim V$ but $\dim V = 5$ and $\dim \text{null } T > 2$ and $\dim \text{range } T = 4$, which is impossible. Observe that since $\dim \text{range } T \geq \dim V$, but $\dim \text{null } T \geq 1$ because T is not injective, then it is impossible for $\dim \text{range } T + \dim \text{null } T$ to equal $\dim V$, as desired.

14 Because T is not surjective, this means that $\dim \text{range } T < \dim W \leq \dim V$ and $\dim \text{null } T > 0$. Then $\dim \text{range } T + \dim \text{null } T < \dim W + \dim \text{null } T \leq \dim V$. Hence, $\dim \text{range } T + \dim \text{null } T < \dim V$, so we're done.

15 Since $v_1, \dots, v_n$ is linearly independent, then since T is injective, we also know that the null space of T equals $\{0\}$, so $Tv_1, \dots, Tv_n$ is also linearly independent because we know that T is a linear map, so $Tv + Tw = T(v + w)$. And since it's injective, each v maps to a distinct output, so we can look at the inputs instead, which are linearly independent.

16 Since T is a linear map from V to W and $v_1, \dots, v_n$ spans V this also means that $Tv_1, \dots, Tv_n$ spans W . $(S_1 S_2 \cdots S_n)(v) = (S_1(v))(S_2(v)) \cdots (S_n(v))$. So if we have

another w such that $(S_1 S_2 \cdots S_n)(w) = (S_1 S_2 \cdots S_n)(v)$, then we must have $S_1(v) = S_1(w)$, etc, so since $S_1, \dots, S_n$ are injective, then $v = w$.

17

- (a) We want to show that T is surjective. Obviously, if T is surjective, then U suffices as the input because we really only need one zero, and the rest would be taken care of if we let U be the set consisting of all inputs that don't result in an output of 0, plus one element from the null set, which is $\{0\}$.
- (b) Observe that the null space consists of points of the form $(5a, a, 7b, b)$, which has dimension 2. Since $\dim \text{null } T + \dim \text{range } T = \dim V = 4$, then $\dim \text{range } T = 2 \dim W$, so T is surjective, as desired. Since $\text{null } T = U$, then since U is 3-dimensional, then $\dim \text{null } T = 3$ as well. $\dim \text{null } T + \dim \text{range } T = \dim V = 8$, so $\dim \text{range } T = 5 = \dim W$, so T is surjective, as desired.

18

- (a) Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. There exists a basis $w_1, \dots, w_n$ of W such that all the entries in the first column of $\mathcal{M}(T)$ (with respect to the bases $v_1, \dots, v_m$ and $w_1, \dots, w_n$) are 0 except for possibly T as a linear map from V to W such that there is nothing except possibly, a 0 in the first row and first column. If we have the matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

In this case, we have a matrix with a 1's along the diagonal, meaning places where the number of rows and number of columns are equal, and 0 everywhere else.

- (b) If $\dim \text{range } T = 1$, this means that $\dim \text{null } T = \dim V - 1$. Therefore there needs to exist a basis of V and a basis of W such that all entries of $\mathcal{M}(T)$ equal 1, because this would We know that $\mathcal{M}(T)$ is defined as the set of numbers $a_{i,j}$ for $1 \leq i \leq m$, $1 \leq j \leq n$ such that for all j ,

$$w_j = a_{1,j}v_1 + a_{2,j}v_2 + \cdots + a_{m,j}v_m.$$

- (c) So, if we add w_{s_j} to w_{t_j} (these are the respective bases of S and T), and let a and b denote the values of the entries in the matrices, we get

$$\mathcal{M}(S + T) = (a_{1,i} + b_{1,i})v_1 + \cdots + (a_{m,i} + b_{m,i})v_m$$

and

$$\begin{aligned} \mathcal{M}(S) + \mathcal{M}(T) &= [a_{1,i}v_1 + \cdots + a_{m,i}v_m] \\ &\quad + [b_{1,i}v_1 + \cdots + b_{m,i}v_m] \\ &= (a_{1,i} + b_{1,i})v_1 + \cdots + (a_{m,i} + b_{m,i})v_m \end{aligned}$$

as desired.

19 If we have

$$Tv_i = a_{1,i}v_1 + \cdots + a_{m,i}v_i,$$

then if we multiply T times λ , we get

$$\lambda a_{1,i}v_1 + \cdots + \lambda a_{m,i}v_i = a_{1,i}(\lambda v_1) + \cdots + a_{m,i}(\lambda v_i).$$

Observe that we wish to prove that when we multiply a matrix by a vector, we get a combination of all of the previous vectors and put them together. This is clear because we already have a collection of vectors to start with. A single value times matrix put together gives a vector of values times matrix.

9.B Operators on Real Inner Product Spaces

1 Since S is an isometry, then $\|Sv\| = \|v\|$. Then $\|S^2v\| = \|Sv\| = \|v\|$, so S^2 is also an isometry. Therefore, there exist an eigenvector x corresponding to eigenvalues 1 and -1 such that $S^2x = x$ since its just satisfying the eigenvalue and there need to be at least 2.

2 It has 1 or -1 as an eigenvalue because $S^2x = x$. Thus $S(Sx) = x$, so basically $S^2 = I$. Hence the eigenvalue is ± 1 .

3 Consider $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and $C = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$. Then

$$AC = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$$

and

$$CA = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$$

as desired.

4 Observe that if $(AB)C$ makes since, then if A is $m \times n$, then B must be $n \times p$ for some p , so AB is $m \times p$, so C is $p \times q$ for some q . Thus BC is $n \times q$, so $A(BC)$ is $m \times q$, which is defined. We already know that the entry in any given row and column of A^2 is equal to the sum of the products of $a_{i,j}, a_{j,i}$, so continuing with A^3 gives $a_{j,p}a_{p,r}a_{r,k}$.

5 Observe that if both T (which is a linear map from U to V) and S which is a linear map from V to W are invertible, then ST , which is a linear map from U to W is invertible, since it's the product of S and T , and thus $ST = T^{-1}S^{-1}$. This doesn't hold because the sum of two noninvertible operators might just be invertible.

6 If there exists an invertible operator $T \in \mathcal{L}(V)$ such that $Tu = Su$ for every $u \in U$, then this must mean that since T is both injective and surjective, then since $Tu = Su$, then so must S . Conversely, if S is injective, then there must exist T such that $Tu = Su$.

7 If $\text{null } T_1 = \text{null } T_2$, this implies that the map from the output of T_2 to the output of T_1 is linear, and it exists, so there exists $S \in \mathcal{L}(W)$ such that $T_1 = ST_2$. If $\text{range } T_1 = \text{range } T_2$, then there exists an invertible operator that's linear and maps from V to V such that $T_1 = T_2S$.

8 Observe that if ST is invertible, then we must be able multiply it by any S^{-1} or T^{-1} , so that we could find S or T . And if S and T are invertible, their determinants are nonzero, so ST 's determinant is also nonzero, so its also invertible.

Chapter 10: Trace and Determinant

Linear Algebra Done Right

Carissa Zeng

October 10, 2024

10.A Trace

1 To prove that $\mathcal{L}(V, W)$ is a vector space, we need to prove that addition and multiplication hold for the set of linear maps. So basically, the set of linear maps is a linear map. Hehe. Anyway consider two linear maps, L_1 and L_2 . Then note that we need to prove that $L_1 + L_2$ and λL_1 are also linear maps.

2 To prove that $L_1 + L_2$ is a linear map, consider two vectors v_1 and v_2 . We want to show that $(L_1 + L_2)(v_1 + v_2) = (L_1 + L_2)v_1 + (L_1 + L_2)v_2$.

$$\begin{aligned}(L_1 + L_2)(v_1 + v_2) &= L_1(v_1 + v_2) + L_2(v_1 + v_2) \\ &= L_1v_1 + L_1v_2 + L_2v_1 + L_2v_2 \\ &= (L_1v_1 + L_2v_1) + (L_1v_2 + L_2v_2) \\ &= (L_1 + L_2)v_1 + (L_1 + L_2)v_2,\end{aligned}$$

since L_1 and L_2 are linear maps.

3 Also,

$$(L_1 + L_2)(\lambda v_1) = L_1(\lambda v_1) + L_2(\lambda v_1) = \lambda L_1v_1 + \lambda L_2v_1 = \lambda(L_1v_1 + L_2v_1) = \lambda(L_1 + L_2)(v_1)$$

so $L_1 + L_2$ is a linear map, as desired.

4 We now show that λL is a linear map.

$$(\lambda L)(v_1 + v_2) = \lambda(L(v_1 + v_2)) = \lambda(Lv_1 + Lv_2) = (\lambda L)v_1 + (\lambda L)v_2$$

so it satisfies summation. Now

$$(\lambda_1 L)(\lambda_2 v) = \lambda_1(L(\lambda_2 v)) = \lambda_1(\lambda_2(Lv)) = \lambda_1\lambda_2(Lv) = \lambda_2((\lambda_1 L)v),$$

as desired. Because L is a linear map itself.

6 Because we know that $\dim V = \dim \text{range } T + \dim \text{null } T$, but $\dim V = 5$, which is odd, so we can't split it evenly. Suppose $S : X \rightarrow Y$ and $T : U \rightarrow V$. Then if we have $x \in X$, which maps to a $y \in Y$, then $y \in \text{null } T$, so $T(y) = T(S(x)) = 0$, so $TS = 0$, so $(ST)^2 = 0$.

7 Observe that since $\dim \text{range } T \geq \dim V$, but $\dim \text{null } T \geq 1$ because T is not injective, then it is impossible for $\dim \text{range } T + \dim \text{null } T$ to equal $\dim V$, as desired.

8 Because T is not surjective, this means that $\dim \text{range } T < \dim W \leq \dim V$ and $\dim \text{null } T$. Then $\dim \text{range } T < \dim W \leq \dim V$. Hence, $\dim \text{null } T + \dim \text{range } T > \dim V$, so we're done.

9 Since $v_1, \dots, v_n$ is linearly independent, then since T is injective, we also know that the null space of T equals 0, so $Tv_1, \dots, Tv_n$ is also linearly independent because we know that T is a linear map, so $Tv + Tw = T(v + w)$. And since its injective, each v maps to a distinct output, so we can look at the inputs instead, which are linearly independent.

10 Since T is a linear map from V to W and $v_1, \dots, v_n$ spans V this also means that $Tv_1, \dots, Tv_n$ spans W .

11 $(S_1 S_2 \cdots S_n)(v) = (S_1(v))(S_2(v)) \cdots (S_n(v))$. So if we have another w such that $(S_1 S_2 \cdots S_n)(w) = (S_1 S_2 \cdots S_n)(v)$, then we must have $S_1(v) = S_1(w)$, etc, so since $S_1, \dots, S_n$ are injective, then $v = w$.

12 We want to show that T is surjective. Obviously, if T is surjective, then U suffices as the input because we really only need one zero, and the rest would be taken care of if we let U be the set consisting of all inputs that don't result in an output of 0, plus one element from the null set, which is $\{0\}$.

13 Observe that the null space consists of points of the form $(5a, a, 7b, b)$, which has dimension 2. Since $\dim \text{null } T + \dim \text{range } T = \dim V = 4$, then $\dim \text{range } T = 2 = \dim W$, so T is surjective, as desired.

14 Since $\text{null } T = U$, then since U is 3-dimensional, then $\dim \text{null } T = 3$ as well. $\dim \text{null } T + \dim \text{range } T = \dim V = 8$, so $\dim \text{range } T = 5 = \dim W$, so T is surjective, as desired.

15 So the null space consists of quintuplets of the form $(3a, a, b, b, b)$, which has dimension 2. Since $\dim \text{range } T = \dim V - \dim \text{null } T = 5 - 2 = 3$, then $\dim \text{range } T > \dim W$, which is impossible.

16 Consider the map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^2$. Then if we map $T(a, b, c, d, e)$ to $(a, b, 0, 0, 0)$, then the range has dimension 2, and the null space consists of all quintuplets of the form $(0, 0, c, d, e)$, which has dimension 3.

17 Suppose $S : X \rightarrow Y$ and $T : U \rightarrow V$. Then if we have $x \in X$, which maps to a $y \in Y$, then $y \in \text{null } T$, so $T(y) = T(S(x)) = 0$, so $TS = 0$, so $(ST)^2 = 0$. Since $(v_1, \dots, v_m)$ encompasses all of V , then the map from the vectors, $\mathbb{F}^m$, to V must be surjective. If $v_1, \dots, v_m$ are linearly independent, this means that the null space has degree 0, and consists of only $\{0\}$.

18 Since T is a linear map from V to W and $v_1, \dots, v_n$ spans V this also means that $Tv_1, \dots, Tv_n$ spans W . Because T is not surjective, this means that $\dim \text{range } T < \dim W \leq \dim V$ and $\dim \text{null } T$. Then $\dim \text{range } T < \dim W \leq \dim V$. Hence, $\dim \text{null } T + \dim \text{range } T > \dim V$, so we're done.

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10.B Determinant

1 Observe that the null space consists of points of the form $(5a, a, 7b, b)$, which has dimension 2. Since $\dim \text{null } T + \dim \text{range } T = \dim V = 4$, then $\dim \text{range } T = 2 = \dim W$, so T is surjective, as desired. Since $\text{null } T = U$, then since U is 3-dimensional, then $\dim \text{null } T = 3$ as well. $\dim \text{null } T + \dim \text{range } T = \dim V = 8$, so $\dim \text{range } T = 5 = \dim W$, so T is surjective, as desired.

2 So the null space consists of quintuplets of the form $(3a, a, b, b, b)$, which has dimension 2. Since $\dim \text{range } T = \dim V - \dim \text{null } T = 5 - 2 = 3$, then $\dim \text{range } T > \dim W$, which is impossible. If $\dim \text{range } T = 1$, this means that $\dim \text{null } T = \dim V - 1$. Therefore there needs to exist a basis of V and a basis of W such that all entries of $\mathcal{M}(T)$ equal 1, because this would imply that they are equal.

3

- (a) Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. There exists a basis $w_1, \dots, w_n$ of W such that all the entries in the first column of $\mathcal{M}(T)$ (with respect to the bases $v_1, \dots, v_m$ and $w_1, \dots, w_n$) are 0 except for possibly T as a linear map from V to W such that there is nothing except possibly, a 0 in the first row and first column.
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$$w_j = a_{1,j}v_1 + a_{2,j}v_2 + \cdots + a_{m,j}v_m.$$

So, if we add w_{s_j} to w_{t_j} (these are the respective bases of S and T), and let a and b denote the values of the entries in the matrices, we get

$$\mathcal{M}(S+T) = (a_{1,i} + b_{1,i})v_1 + \cdots + (a_{m,i} + b_{m,i})v_m$$

and

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as desired.

4 If we have

$$Tv_i = a_{1,i}v_1 + \cdots + a_{m,i}v_i,$$

then if we multiply T times λ , we get

$$\lambda a_{1,i}v_1 + \cdots + \lambda a_{m,i}v_i = a_{1,i}(\lambda v_1) + \cdots + a_{m,i}(\lambda v_i)$$

as desired. Observe that we wish to prove that when we multiply a matrix by a vector, we get a combination of all of the previous vectors and put them together. This is clear because we already have a collection of vectors to start with. A single value times matrix put together gives a vector of values times matrix.

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8 Observe that if ST is invertible, then we must be able multiply it by any S^{-1} or T^{-1} , so that we could find S or T . And if S and T are invertible, their determinants are nonzero, so ST 's determinant is also nonzero, so it's also invertible. If $ST = I$, then ST is invertible, so $(ST)^{-1} = T^{-1}S^{-1} = I$, so multiplying by T on the left gives $S^{-1} = T$, so multiplying by S on the right gives $TS = I$.

9 If $STU = I$, multiplying by S^{-1} on the left gives $TU = S^{-1}$, then multiplying by U^{-1} on the right gives $T = S^{-1}U^{-1}$, which we know equals $(US)^{-1}$ by problem 1, so $T^{-1} = US$. If V isn't finite-dimensional, then S , T , and U might not necessarily have inverses, so there goes the proof.

10 If R , S , and T are all linear maps from V to V itself, since RST is plugging the output of T into S and plugging this into R , then since T is surjective, then S is injective. It's a linear map from $\mathbb{F}$ to V . Since $\dim \mathbb{F} = 1$, then it's isomorphic. Since $\deg Tp \leq \deg p$ for all p , this must imply that T is a bijection (since it's also injective), so it must be surjective.

11 Observe that if the only solution if $c_1 = \dots = c_n = 0$ is all 0's, then this means that solutions exist, and it's linearly independent ($A_{.,k}$ for all k) so it can also act as a basis, so there will have to exist x_k . And if there exist x_k as solutions to the c , then it's a basis, so it's linearly independent, so $x_1 = \dots = x_n = 0$ is the only solution if all the c 's equal 0. Since it's surjective, then $\deg Tp = \deg p$ otherwise it wouldn't be surjective. Tada.

12 We want to find the volume of the ellipsoid with the semi-axes lengths being a , b , and c . Consider the set

$$\Omega = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 < a^3\},$$

which is basically the sphere with radius a . Thus $T(\Omega) = \frac{4}{3}\pi a^3$, so by scaling $T(\Omega)$ by $\frac{b}{a}$ and $\frac{c}{a}$ for the side lengths of the ellipsoid to get $\frac{4}{3}\pi abc$.